CHAPTER 53

Chest wall and breast

The chest wall surrounds the thoracic cavity. The skin and soft tissue cover a musculoskeletal frame consisting of twelve pairs of ribs, which articulate with twelve thoracic vertebrae posteriorly and (except for the last two pairs of ribs) with the sternum anteriorly, via their costal cartilages; intrinsic muscles and muscles that connect the chest wall with the upper limb and the vertebral column; and numerous blood and lymphatic vessels and nerves that supply the components of the musculoskeletal frame and the overlying skin and breast tissue.

SKIN AND SOFT TISSUE

SKIN

Vascular supply

Arteries

The skin of the thorax is supplied by a combination of direct cutaneous vessels and musculocutaneous perforators that reach the skin primarily via the intercostal muscles, pectoralis major, latissimus dorsi and trapezius. The major contributing vessels are branches from the thoraco-acromial axis, lateral and internal thoracic, anterior and posterior intercostal, thoracodorsal, transverse cervical, dorsal scapular and circumflex scapular arteries (**Figs 53.1**–**53.2**; see Fig. 43.4).

The anterior aspect of the thoracic skin is supplied by the thoraco-acromial axis, the internal thoracic arteries, perforating branches from the intercostal arteries and branches from the lateral thoracic and superficial thoracic arteries. The thoraco-acromial axis supplies the skin primarily via musculocutaneous perforators from its pectoral branch; they reach the skin through pectoralis major. In addition, direct cutaneous branches arise from the acromial and deltoid branches. The internal thoracic artery sends direct perforating branches to the skin of the upper six intercostal spaces, accompanied by the cutaneous branches of the anterior intercostal nerves. The branches reach the skin after passing through pectoralis major and travelling laterally in the subcutaneous fat as direct cutaneous vessels. The second intercostal perforator is usually the largest. The lateral aspect of the thoracic skin is supplied by the lateral thoracic and superficial thoracic arteries and by lateral cutaneous branches of the intercostal arteries. The lateral thoracic artery gives off direct cutaneous branches to the lateral chest wall in addition to musculocutaneous branches that pass through pectoralis major. The posterior aspect of the thoracic skin is supplied by the medial and lateral dorsal cutaneous branches of the posterior intercostal arteries (via erector spinae and latissimus dorsi); musculocutaneous perforating branches from the superficial cervical, transverse cervical and dorsal scapular arteries (via trapezius); musculocutaneous perforating branches from the thoracodorsal artery and the intercostal arteries (via latissimus dorsi); and direct cutaneous branches from the circumflex scapular artery.

Veins

The intercostal veins accompany the similarly named arteries in the intercostal spaces (see Fig. 51.3). The small anterior intercostal veins are tributaries of the musculophrenic and internal thoracic veins, the latter draining into the appropriate brachiocephalic vein. The posterior intercostal veins drain backwards directly or indirectly into the azygos vein on the right and the hemiazygos or accessory hemiazygos veins on the left. The azygos veins exhibit great variation in their origin, course, tributaries, anastomoses and termination (Ch. 56).

Lymphatic drainage

Superficial lymphatic vessels of the thoracic wall ramify subcutaneously and converge on the axillary nodes (p. 834; see Figs 53.22, 53.24, 48.45). Lymph vessels from the deeper tissues of the thoracic walls drain mainly to the parasternal, intercostal and diaphragmatic lymph nodes.

Innervation

The skin of the thorax is supplied by cutaneous branches of cervical and thoracic nerves in consecutive, curved zones; the upper zones are almost horizontal and the lower are oblique. On the upper anterior aspect of the thorax, the areas innervated by the third and fourth cervical nerves adjoin those innervated by the first and second thoracic nerves because the intervening nerves provide the sensory and motor supply to the upper limb (Ladak et al 2014) (**Fig. 53.3**). There is a similar, but less extensive, posterior 'gap': most of the skin covering the posterior aspect of the thorax is supplied by the dorsal rami of the thoracic nerves (see Fig. 45.8). The subcostal margin is supplied by the seventh thoracic nerve.

The ventral rami of the first to the eleventh thoracic nerves pass into the appropriate intercostal space, giving off a lateral cutaneous branch that arises beyond the angle of the rib and divides into anterior and posterior branches, terminating near the sternum as an anterior cutaneous branch (see Fig. 53.21).

Branches of the supraclavicular nerve, which originates from the third and fourth cervical nerve roots, supply the skin in the upper pectoral region. Most of the first thoracic nerve joins the brachial plexus, apart from a small inferior branch that becomes the first intercostal nerve. The lateral cutaneous branch of the second intercostal nerve supplies the skin of the axilla as the intercostobrachial nerve. The costal margin is supplied by a branch from the seventh thoracic nerve. The seventh to eleventh thoracic nerves supply the skin of the thoracic wall as they pass anteriorly and inferiorly; they continue beyond the costal cartilages to supply the skin and subcutaneous tissues of the abdominal wall. The skin of the abdomen at the level of the umbilicus is supplied by the tenth thoracic nerve. The subcostal nerve follows the inferior border of the twelfth rib and supplies the skin of the lower abdominal wall (see Fig. 61.5).

SOFT TISSUE

Superficial fascia

The superficial fascia consists primarily of fat and is only loosely attached to the skin, an arrangement that allows some movement of the underlying structures. Small blood vessels and nerves perforate the superficial fascia to supply the skin. The breast lies within the superficial fascia, apart from a superolateral extension that pierces the deep fascia to form the axillary tail in the female. It is described later in this chapter.

Deep fascia

Clavipectoral fascia

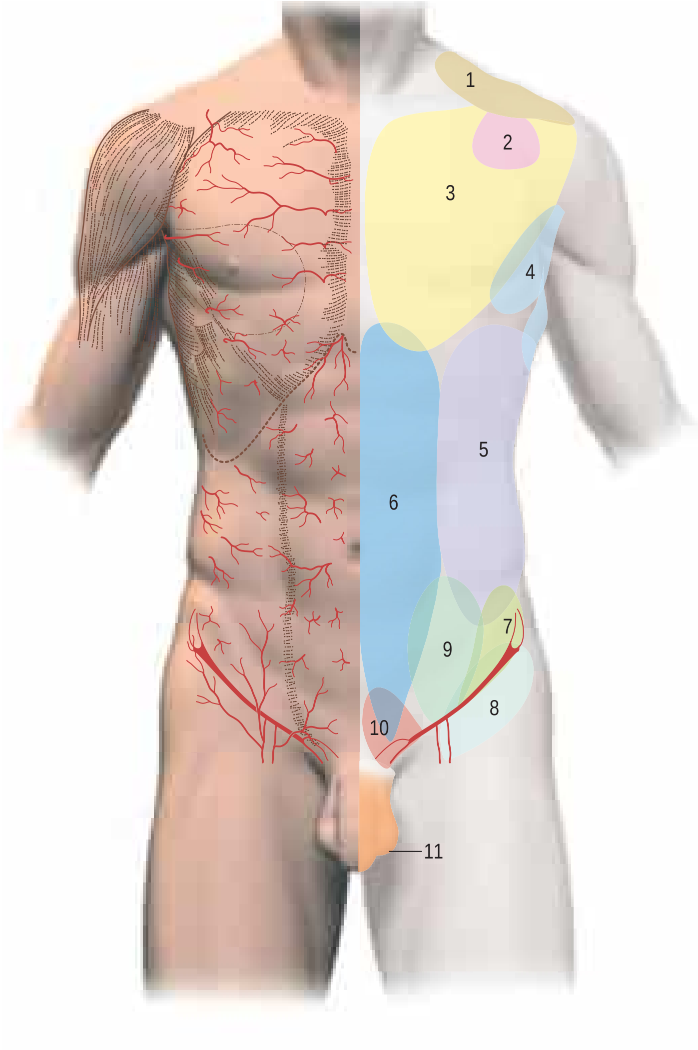
The clavipectoral fascia is the cranial continuation of the deep lamina of the pectoral fascia and the medial continuation of the parietal layer of the subscapular bursal fascia. It extends between the clavicle and pectoralis minor, surrounds subclavius and extends medially to the first rib. It is described in detail on page 799.

BONE AND CARTILAGE

The twelve thoracic vertebrae and their associated intervertebral discs are described in detail in Chapter 43.

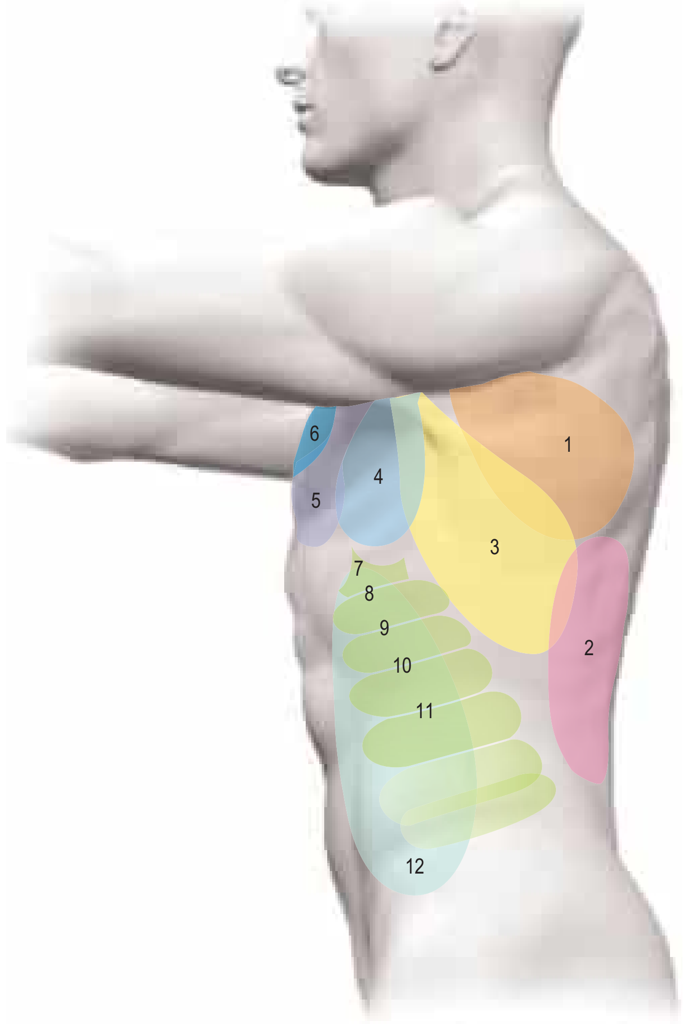
STERNUM

The sternum is an elongated, flattened bone that forms the middle portion of the anterior wall of the thorax. It articulates with the clavicles



- 1 Transverse cervical artery
- 2 Direct cutaneous branch of the thoraco-acromial axis
- 3 Anterior perforators from the interior thoracic artery
- 4 Superficial thoracic artery
- 5 Intercostal perforators
- 6 Musculocutaneous perforators from the superior and inferior deep epigastric arteries
- 7 Deep circumflex iliac artery
- 8 Superficial circumflex iliac artery
- 9 Superficial inferior epigastric artery
- 10 Superficial external pudendal artery
- 11 Deep external pudendal artery

Fig. 53.1 Anatomical territories of cutaneous blood vessels on the anterior trunk. (With permission from Cormack GC, Lamberty BGH 1994 The Arterial Anatomy of Skin Flaps, 2nd ed. Edinburgh: Churchill Livingstone.)



- 1 Circumflex scapular artery
- 2 Musculocutaneous perforators through latissimus dorsi from intercostal and lumbar arteries
- 3 Musculocutaneous perforators through latissimus dorsi from thoracodorsal artery
- 4 Cutaneous branches from thoracodorsal and lateral thoracic arteries
- 5 Thoraco-acromial axis
- 6 Internal thoracic artery
- 7–11 Lateral cutaneous branches from intercostal arteries
- 12 Musculocutaneous perforators through external oblique from intercostal arteries

Fig. 53.2 Anatomical territories of cutaneous blood vessels on the lateral trunk. (With permission from Cormack GC, Lamberty BGH 1994 The Arterial Anatomy of Skin Flaps, 2nd ed. Edinburgh: Churchill Livingstone.)

at the sternoclavicular joints (p. 808) and with the cartilages of the first seven pairs of ribs. The sternum consists of a cranial manubrium, an intermediate body (mesosternum, gladiolus) and a caudal xiphoid process (Figs 53.4–53.5). Until puberty, the body consists of four sternebrae, which are intersegmental. The average length of the adult sternum is 17 cm; the ratio between manubrial and mesosternal lengths differs between the sexes. Growth may continue beyond the third decade and possibly throughout life.

In natural stance, the sternum slopes down and slightly forwards. The bone is anteriorly convex and posteriorly concave, and broadest at the junction with the first costal cartilages. It narrows at the manubriosternal joint, widens to its articulation with the fifth costal cartilages, and then narrows again. The sternum contains highly vascular trabecular bone enclosed by a compact layer that is thickest in the manubrium between the clavicular notches. Centrally, the bone is lightly constructed whereas, laterally, the trabeculae are thicker and wider. The medulla contains haemopoietic bone marrow.

Manubrium

The manubrium is level with the third and fourth thoracic vertebrae. It has a somewhat quadrangular form, being broad and thick above and narrowing to its junction with the body. The anterior surface is smooth, transversely convex and vertically concave, and the posterior surface is concave and smooth. The superior border is thick and contains a central jugular (suprasternal) notch between two oval fossae, the clavicular notches, that are directed up and posterolaterally for articulation with the sternal ends of the clavicles. Fibres of the interclavicular ligament are attached to the jugular notch. The inferior border, oval and rough, carries a thin layer of cartilage for articulation with the body. The lateral borders are marked above by a depression for the first costal cartilage and below by a small articular demifacet, which articulates with part of the second costal cartilage. The narrow curved edge descends medially between these facets. There is very little accessory movement at the manubriosternal joint, although pressure on either the body or the manubrium close to the joint will produce slight angling of the two sections.

Unlike all the other sternocostal joints, the joint between the manubrium and the first costal cartilage is a fibrous synarthrosis. It lies 1 cm below and 1 cm lateral to the medial end of the clavicle and is difficult to palpate.

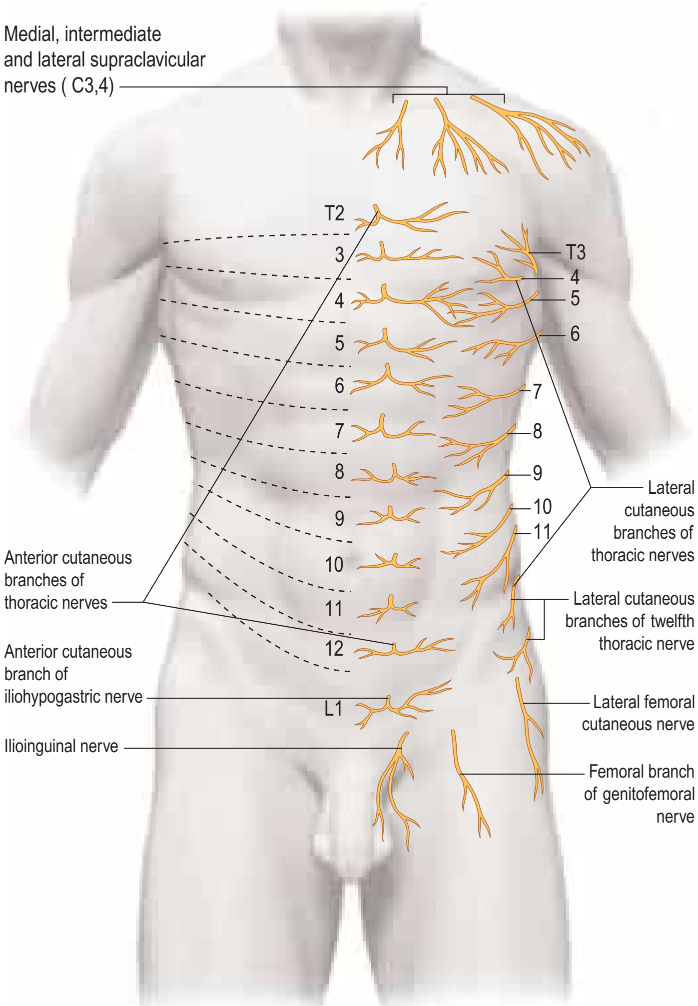


Fig. 53.3 The approximate segmental distribution of the cutaneous nerves on the anterior trunk. The contribution from the first thoracic spinal nerve is not shown and the considerable overlap that occurs between adjacent segments is not indicated.

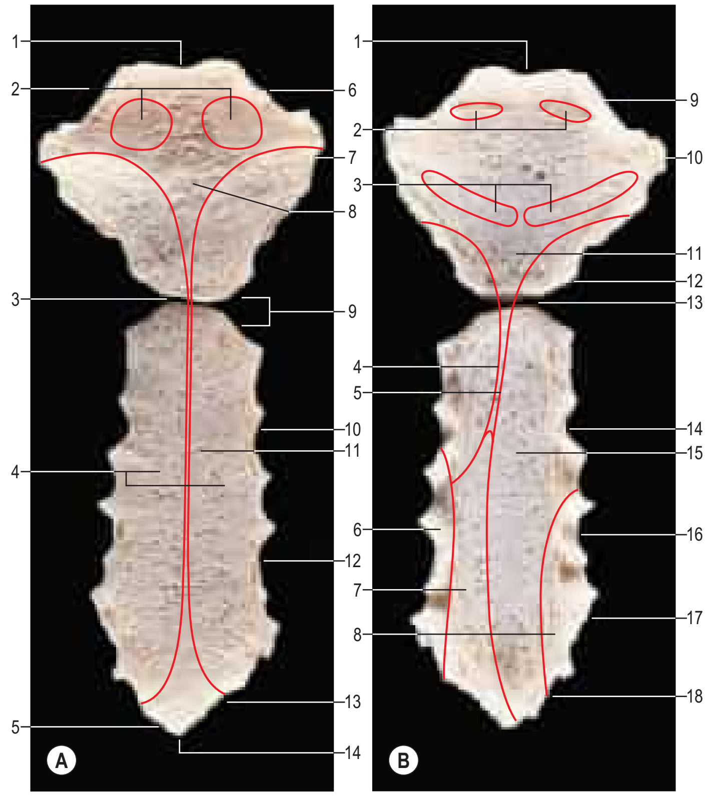


Fig. 53.4 The sternum. **A,** Anterior aspect. Key: 1, jugular notch; 2, attachment for sternocleidomastoid; 3, sternal angle and manubriosternal joint; 4, attachment for pectoralis major; 5, seventh costal notch; 6, clavicular notch; 7, first costal notch 8, manubrium; 9, second costal notch; 10, third costal notch; 11, body of sternum; 12, fourth costal notch; 13, sixth costal notch; 14, xiphisternal joint. **B,** Posterior aspect. Key: 1, jugular notch; 2, attachment for sternohyoid; 3, attachment for sternothyroid; 4, edge of area covered by left pleura; 5, edge of area covered by right pleura; 6, attachment for transversus thoracis; 7, area in contact with pericardium; 8, attachment for transversus thoracis; 9, clavicular notch; 10, first costal notch; 11, manubrium; 12, second costal notch; 13, sternal angle and manubriosternal joint; 14, third costal notch; 15, body of sternum; 16, fourth costal notch; 17, fifth costal notch; 18, seventh costal notch.

Body

The body is level with the fifth to ninth thoracic vertebrae. It is longer, narrower and thinner than the manubrium, and is broadest near its lower end. The anterior surface, being nearly flat and facing slightly upwards, usually bears three variable transverse ridges that mark the levels of fusion of its four sternebrae. A sternal foramen, of varying size and form, may occur between the third and fourth sternebrae. The posterior surface, slightly concave, also displays three less distinct transverse lines. The oval upper end articulates with the manubrium at the level of the sternal angle (manubriosternal joint, angle of Louis), which lies opposite the inferior border of the fourth vertebral body and is marked by a posterior transverse groove.

Named after the French surgeon, Antoine Louis (1723–1792), the Latin name, angulus Ludovici, is not infrequently mistranslated into English as 'the angle of Ludwig'.

The lower end of the body is narrow and continuous with the xiphoid process. On each lateral border, at its superior angle, a small notch articulates with part of the second costal cartilage (see Figs 53.4, 53.5). Below this, four costal notches articulate with the third to sixth costal cartilages. The inferior angle bears a small facet, which, together with the xiphoid process, articulates with the seventh costal cartilage. Between these articular depressions, a series of curved edges diminish in length downwards and form the anterior limits of the intercostal spaces.

Xiphoid process (xiphisternum)

The xiphoid process is in the epigastrium. It is the smallest and most variable sternal element, and may be broad and thin, pointed, bifid, perforated, curved or deflected. When elongated and curved forwards, it may be mistaken for an epigastric mass. The xiphoid is cartilaginous in youth but more or less ossified in adults. It is continuous with the lower end of the body at the xiphisternal joint. Demifacets that articulate with parts of the seventh costal cartilages occur anterior to its superolateral angles (see Fig. 53.5).

Muscle attachments The sternal ends of pectoralis major and sternocleidomastoid are attached to the anterior surface of the manubrium. Sternothyroid is attached to the posterior surface, opposite the first costal cartilage, and the most medial fibres of sternohyoid are attached above sternothyroid. The articular capsules of the sternocostal joints and sternal fibres of pectoralis major are attached to the anterior surface of the body. Transversus thoracis (sternocostalis) is attached to its posterior surface. The external intercostal membranes are attached to the borders of the body between the costal facets. The most medial fibres of rectus abdominis and the aponeuroses of external and internal oblique are attached to the anterior surface of the xiphoid. The linea alba is attached to its lower end, and the aponeuroses of internal oblique and transversus abdominis are attached to its borders. Slips of the diaphragm are attached to its posterior aspect, and the sternum is here related to the liver.

Vascular supply The internal thoracic artery provides the main blood supply for the sternum via anterior and posterior networks of perforating arteries that directly penetrate the sternum at the level of each intercostal space. These networks are particularly well developed posteriorly and at the level of the fourth and fifth intercostal spaces.

The venous network is less developed. It is formed mainly by an inframedullary network of sinuses in the bone that drain via transcortical veins either into the peripheral sternal networks or the internal thoracic vein.

Innervation The manubrium is supplied by the anterior branch of the supraclavicular nerve and the anterior cutaneous branch of the first intercostal nerve, whereas the body of the sternum is supplied largely by anterior branches of the intercostal nerves. The anterior branch of the phrenic nerve runs anteromedially from the diaphragm and supplies the lower portion of the sternum.

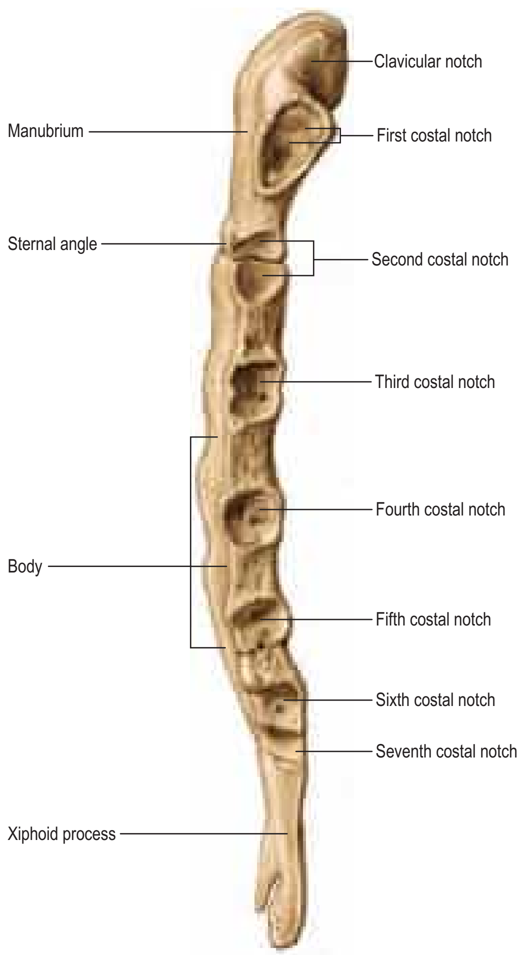


Fig. 53.5 The sternum, left lateral aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer, 2013.)

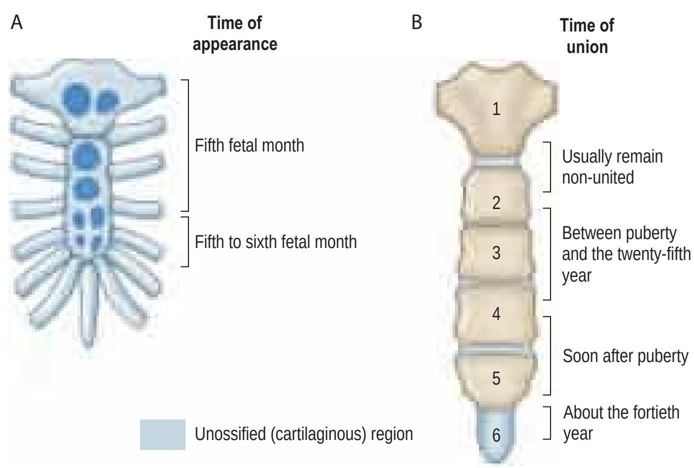


Fig. 53.6 The ossification of the sternum. **A,** Before birth. **B,** After birth.

Ossification The sternum is formed by fusion of two cartilaginous sternal plates flanking the median plane. The arrangement and number of centres of ossification vary according to the level of completeness and time of fusion of the sternal plates, and according to the width of the adult bone. Incomplete fusion leaves a sternal foramen. The manubrium is ossified from one to three centres appearing in the fifth fetal month. The first and second sternebrae usually ossify from single centres that appear at about the same time (Fig. 53.6A). Centres in the third and fourth sternebrae are commonly paired, and appear in the fifth and sixth months, respectively; one of either pair may be delayed until the seventh or even eighth month, and the fourth sternebral centre may be absent. The xiphoid process begins to ossify in the third year or later. In some sterna, all centres are single and median; in others, the manubrial centre is single and the sternebral centres are all paired, symmetric or asymmetric. Union between mesosternal centres begins at puberty and proceeds from below upwards; by the age of 25 years, they are all united (Fig. 53.6B).

Suprasternal ossicles, paired or single, sometimes occur. They may fuse to the manubrium or articulate posteriorly at the lateral border of the jugular notch. When well formed, they are pyramidal, and their base is articular. The ossicles are cartilaginous at birth and ossify during adolescence.

Pectus excavatum Pectus excavatum is the most common congenital deformity of the anterior chest, and is a depression of the sternum and costal cartilage that is sometimes referred to as cobbler's chest, sunken chest, the crevasse or funnel chest. The lower costal cartilages and the body of the sternum are depressed and there is some asymmetric curving of the ribs posteriorly. There is often an abnormal posture with dorsal lordosis, sometimes with developing scoliosis. The deformity is found either at birth (1 in 500 live births) or early in life; in the majority of cases, the condition is manifest by 1 year but may not develop until puberty. The aetiology is unknown but some cases are associated with connective tissues diseases, such as Marfan's syndrome.

Pectus carinatum Pectus carinatum, often called 'pigeon chest', is an overgrowth of cartilage that causes the sternum to protrude forwards. It is less common than pectus excavatum and a quarter of cases are familial. The condition may occur as a solitary congenital abnormality or in association with other genetic disorders. Although present in early childhood, pectus carinatum usually progresses during adolescence and then remains unchanged throughout adulthood.

CLAVICLE

The clavicle is described in Chapter 48.

RIBS

The ribs are 12 pairs of elastic arches (Fig. 53.8), each consisting of highly vascular trabecular bone containing large amounts of red marrow, enclosed in a thin layer of compact bone.

The ribs articulate posteriorly with the vertebral column and form the greater part of the thoracic skeleton. Their number may be increased by cervical or lumbar ribs or reduced by the absence of the twelfth pair. The first seven 'true' ribs connect to the sternum by costal cartilages, whilst the remaining lower five 'false' ribs either join the superjacent costal cartilage (8–10) or 'float' free at their anterior ends as relatively small and delicate structures tipped with cartilage (11–12). The tenth rib may also float: the incidence varies from 35% to 70%, depending on ancestry.

The ribs are separated by the intercostal spaces, which are deeper in front and between the upper ribs. The latter are less oblique than the lower ribs; obliquity is maximal at the ninth rib. Ribs increase in length up to the seventh and thereafter diminish. They decrease in breadth downwards; in the upper ten the greatest breadth is anterior. The first two and the last three ribs present special features, whereas the remainder conform to a common plan.

Typical rib

A typical rib has a shaft with anterior and posterior ends (Fig. 53.9). The anterior costal end has a small concave depression for the lateral end of its cartilage. The shaft has an external convexity and is grooved internally near its lower border, which is sharp, whereas its upper border is rounded. The posterior vertebral end has a head, neck and tubercle. The head presents two facets, separated by a transverse crest. The lower and larger facet articulates with the body of the corresponding vertebra, its crest attaching to the intervertebral disc above it. The neck is the flat part beyond the head, anterior to the corresponding transverse process. It is oblique and faces anterosuperiorly. Its posteroinferior surface is rough and pierced by foramina. Its upper border is the sharp crest of the neck, its lower border rounded. The tubercle, which is more prominent in upper ribs, is posteroexternal at the junction of the neck and shaft, and is divided into medial articular and lateral non-articular areas. The articular part bears a small, oval facet for the transverse process of the corresponding vertebra. The non-articular area is roughened by ligaments. The shaft is thin and flat, and has external and internal surfaces, and superior and inferior borders. It is curved, bent at the posterior angle (5–6 cm from the tubercle), and twisted about its long axis. The part behind the angle inclines superomedially and so its external surface is posteroinferior. In front of the angle, the rib faces slightly up. It is convex and smooth, and, near the

The clinical significance of pectus excavatum may vary considerably (**Fig. 53.7**). Severe defects can lead to cardiopulmonary dysfunction, whilst almost all patients will have some degree of cosmetic concern. Back and chest pain may occur. Evaluation of the deformity demands careful symptom assessment in addition to baseline radiological investigations (chest radiograph and computed tomography (CT)). Where there are physiological restrictions, lung function tests, electrocardiograms (ECGs) and echocardiograms form part of the assessment before specialist clinicians can determine the safety of any surgical intervention. A classification system based on morphology using CT scanning has been proposed (Park et al 2004) and is useful for standardizing the results of surgical correction of the different subtypes.

Mild defects causing aesthetic concerns may be improved with customized silicone implants (Masson et al 1970) or with soft tissue reconstructive procedures (Raab et al 2009), to camouflage the concavity. More severe defects often require correction of the underlying skeletal deformity. Several techniques have been described. Open resection procedures were pioneered by Ravitch; however, more recently, minimally invasive techniques have gained in popularity (Ravitch 1955, Nuss et al 1988).

Pectus carinatum occurs in three different ways, most commonly in the 11–14-year-old pubertal male undergoing a growth spurt. Some parents report that their child's pectus seemingly popped up 'overnight'. The second most common occurrence is from birth, when it is evident in newborns as a rounded chest, becoming more prominent as the child reaches 2 or 3 years old. The least common occurrence is as an acquired condition after open heart surgery, when healing has been aberrant. Pectus carinatum may be associated with Turner's, Marfan's, Ehlers–Danlos, Morquio's, Noonan's, Sly's or multiple lentigines syndromes, trisomies 18 or 21, homocystinuria, osteogenesis imperfecta or scoliosis.

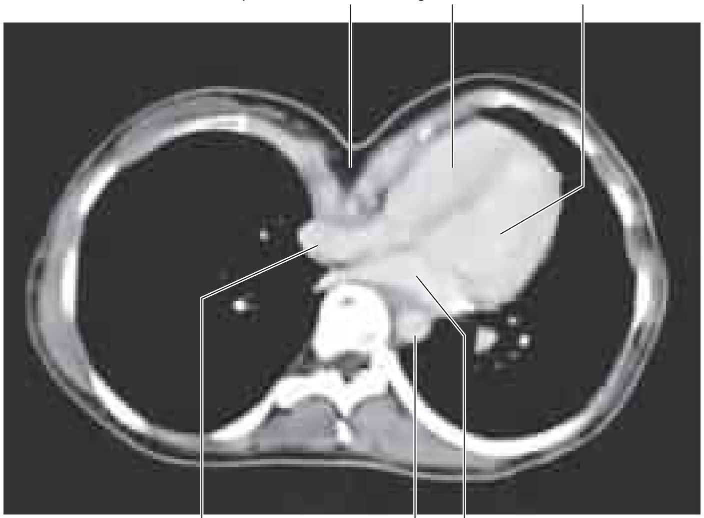


Fig. 53.7 Pectus excavatum.

CHAPTER 53

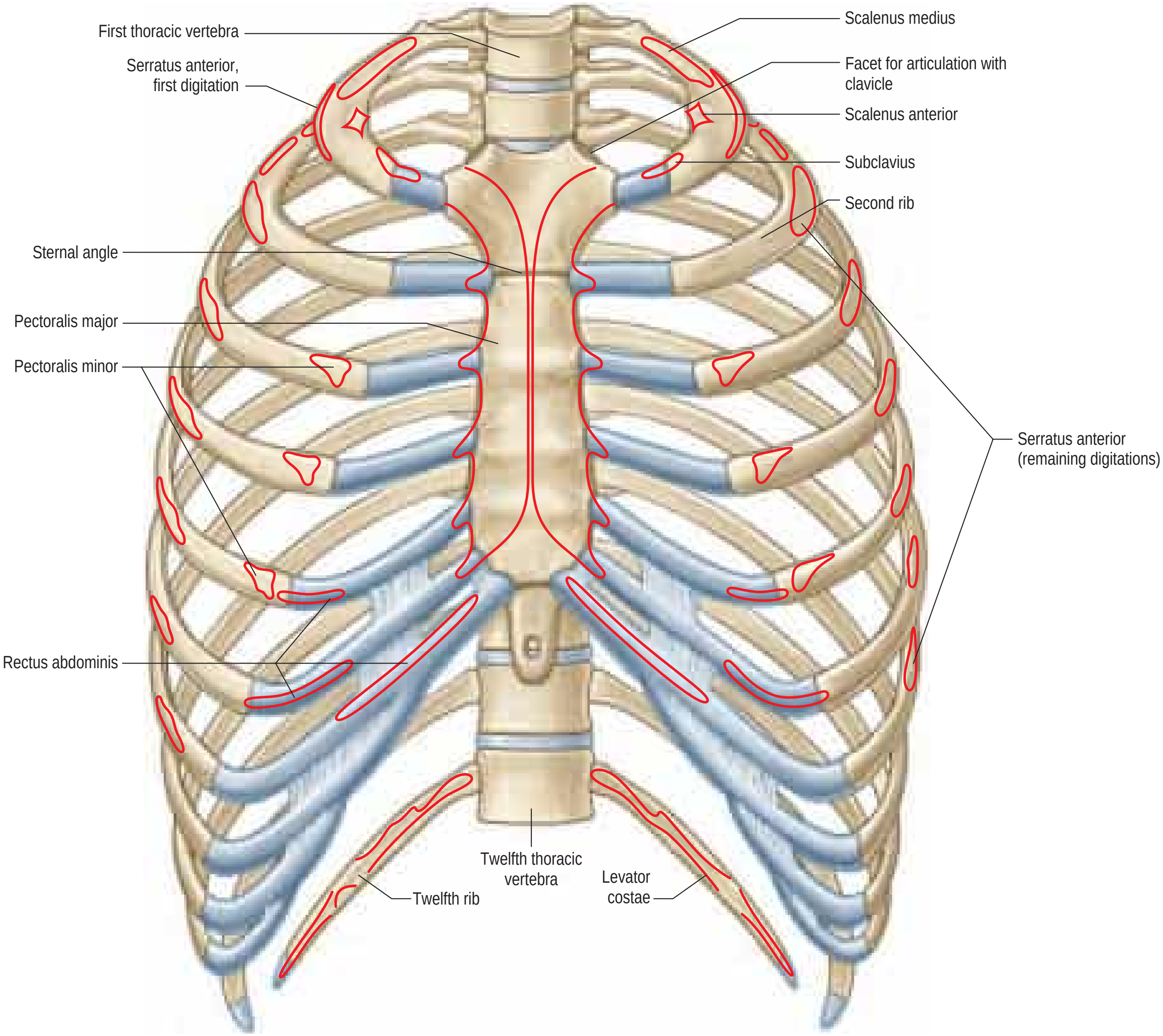


Fig. 53.8 The skeleton of the thorax, anterior aspect, showing muscle attachments.

tubercle, is crossed by a rough line, directed inferolaterally, towards the posterior angle. The smooth internal surface is marked by a costal groove, bounded below by the inferior border. The superior border of the groove continues behind the lower border of the neck, but terminates anteriorly at the junction of the middle and anterior thirds of the shaft, anterior to which the groove is absent.

Attachments and relations A radiate ligament is attached along the anterior border of the head and an intra-articular ligament is attached along its crest. The anterior surface of the head is related to costal pleura and, in the more inferior ribs, to the sympathetic trunk. The anterior surface of the neck is divided by a faint transverse ridge for the internal intercostal membrane and is continuous with the inner lip of the superior border of the shaft. The area above the ridge, which is more or less triangular, is separated from the membrane by fatty tissue whilst the inferior smooth area is covered by costal pleura. The posterior surface of the neck gives attachment to the costotransverse ligament and is pierced by vascular foramina. The superior costotransverse ligament is attached to the crest of the neck, which extends laterally into the outer lip of the superior border of the shaft. The rounded inferior border of the neck continues laterally into the upper border of the costal groove, and gives attachment to the internal intercostal membrane. The articular area of the tubercle in the upper six ribs is convex and faces posteromedially. In the succeeding three or four ribs it is almost flat, and faces down, back and slightly medially. The lateral costotransverse ligament is attached to the non-articular area.

The ridge on the external surface of the shaft (near its posterior angle) gives attachment to an upward continuation of the thoracolumbar fascia and lateral fibres of iliocostalis thoracis. From the second to the tenth ribs, the distance between angle and tubercle increases. Medial to the angle, the external surface gives attachment to levator costae and is covered by erector spinae. Near the sternal end of this surface, an indistinct oblique line, the anterior 'angle', separates the attachments of external oblique and serratus anterior (or latissimus dorsi, in the case of the ninth and tenth ribs). The internal intercostal muscle is attached to the costal groove on the internal surface, and separates the bone and the intercostal neurovascular bundle. At its vertebral end, the groove faces down, its borders in the same plane. The shaft broadens near the posterior angle, and the groove reaches its internal surface. The innermost intercostal is attached to the superior rim of the groove, and this attachment occasionally extends to the anterior quarter of the rib. Posteriorly, the superior rim meets the lower border of the neck. The external intercostal muscle is attached to the sharp inferior costal border. The superior border has two lips posteriorly: an inner and an outer lip. The internal intercostal muscles and the innermost intercostal muscles are attached to the inner lip. The external intercostal muscle is attached to the outer lip.

Vascular supply and innervation Typical ribs receive their blood supply anteriorly via branches from the internal thoracic artery (first six intercostal spaces) or musculophrenic artery (subsequent spaces), and posteriorly from intercostal arteries derived directly from the aorta. Venous drainage is into the corresponding intercostal vein and thence into the azygos system. Typical ribs are innervated segmentally by branches from their corresponding intercostal nerves.

Cervical rib

A cervical rib, the costal element of the seventh cervical vertebra, may be a mere epiphysis on its transverse process but more often it has a head, neck and tubercle. When a shaft is present, it is of variable length and extends anterolaterally into the posterior triangle of the neck, where it may end freely or join the first rib, its costal cartilage or even the sternum. A cervical rib may be partly fibrous but its effects are not related to the size of its osseous part. If it is long enough, its relations are those of a first thoracic rib; the brachial plexus (usually lower trunk)

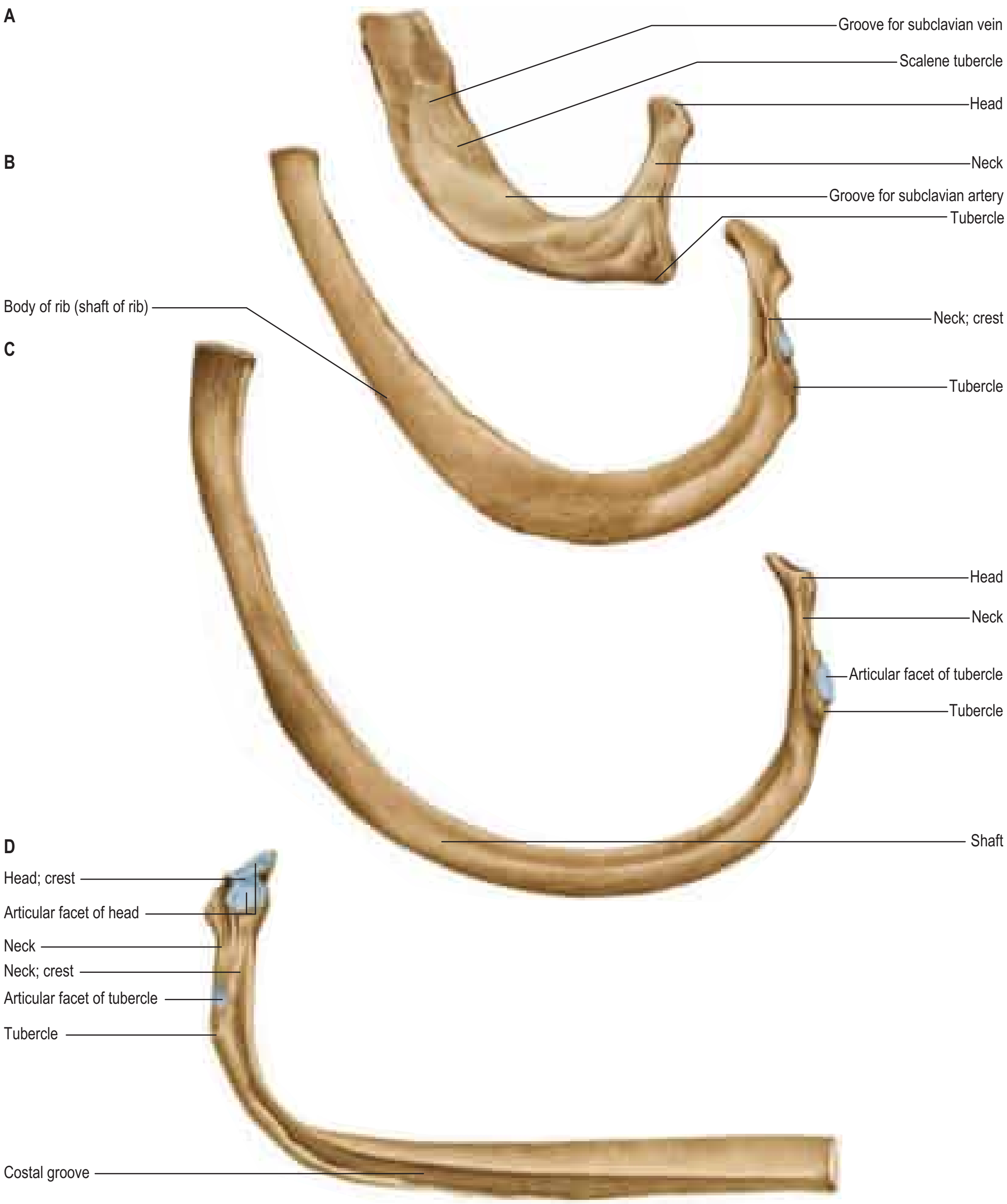


Fig. 53.9 Superior surfaces of selected ribs, left side. **A,** First rib. **B,** Second rib. **C,** Third rib. **D,** Inferior surface of eighth rib, left side. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer, 2013.)

and subclavian vessels are superior and apt to suffer compression in a narrow angle between the rib and scalenus anterior. Hence, cervical ribs may first be revealed by neurovascular symptoms, particularly those caused by pressure on the eighth cervical and first thoracic spinal nerves.

A cervical rib (pleurapophysis) may show synostosis or diarthrosis with either the anterior (parapophysial) or posterior (diapophysial) 'roots' of the so-called seventh cervical transverse process or, more usually, with both.

First rib

Most acutely curved and usually shortest, the first rib is broad and flat, its surfaces are superior and inferior, and its borders are internal and external (see Fig. 53.9A). It slopes obliquely down and forwards to its sternal end. The obliquity of the first rib accounts for the ingress of the pulmonary and pleural apices into the neck.

The head of the first rib is small and round, and bears an almost circular facet that articulates with the body of the first thoracic vertebra. The neck is rounded and ascends posterolaterally, and the tubercle, wide and prominent, is directed up and backwards. Medially, an oval facet articulates with the transverse process of the first thoracic vertebra. At the tubercle, the rib is bent, its head turned slightly down, and so the angle and tubercle coincide. The superior surface of the flattened shaft is crossed obliquely by two shallow grooves, separated by a slight ridge, which usually ends at the internal border as a small pointed projection, the scalene tubercle, to which scalenus anterior is attached. The groove anterior to the scalene tubercle forms a bed for the subclavian vein, and the rough area between this and the first costal cartilage gives attachment to the costoclavicular ligament and, more anteriorly, to subclavius. The subclavian artery and (usually) the lower trunk of the brachial plexus pass in the groove behind the tubercle. Behind this, scalenus medius is attached as far as the costal tubercle.

The external border is convex, thick posteriorly and thin anteriorly. It is covered behind by scalenus posterior descending to the second rib. The first digitation of serratus anterior is, in part, attached to it, behind the subclavian (arterial) groove. The internal border is concave and thin, and the scalene tubercle is near its midpoint. The suprapleural membrane, which covers the cervical dome of the pleura, is attached to the internal border. The inferior surface is smooth and the anterior end is larger than in any other rib.

Vascular supply and innervation The first rib is supplied by the internal thoracic and superior intercostal arteries, drained by the intercostal vein and innervated by the first intercostal nerve.

Ossification The first rib has a primary centre for the shaft, and secondary ossification centres for the head of the rib and the tubercle.

Second rib

The second rib is twice the length of the first and has a similar curvature. The non-articular area of its tubercle is small. The angle is slight and near the tubercle. The shaft is not twisted but at the tubercle is convex upwards, as in the first rib but less so. The external surface of the shaft is convex and superolaterally is marked centrally by a rough, muscular impression that continues posteromedially towards the tubercle as a narrow, roughened ridge. The internal surface, smooth and concave, faces inferomedially and there is a short costal groove posteriorly.

The lower parts of the first two digitations of serratus anterior are attached to a rough prominence that extends from just behind the midpoint of the external surface (see Fig. 53.9B). The distinct lips of the upper border are widely separated behind; scalenus posterior and serratus posterior superior are attached to the outer lip in front of the angle.

Vascular supply and innervation The blood supply of the second rib is via the internal thoracic and superior intercostal arteries. Venous drainage is via the superior intercostal vein, which drains into the brachiocephalic vein, and the anterior intercostal veins, which drain into the internal thoracic vein. The bone is innervated by branches of the first intercostal nerve.

Ossification The second rib is ossified from a primary centre for the shaft, which appears near the angle late in the second month. The secondary centres for the head and articular and non-articular parts of the tubercle appear about puberty, uniting to the shaft soon after the age of 20 years.

Tenth, eleventh and twelfth ribs

The tenth rib has a single facet on its head that may articulate with the intervertebral disc above, in addition to the upper border of the tenth thoracic vertebra near its pedicle. The ninth and tenth ribs are usually united anteriorly by a fibrous joint. However, the tenth rib may be free in 35–70% (depending on ancestry), in which case it is pointed like the eleventh and twelfth ribs.

The eleventh and twelfth ribs each have one large articular facet on the head but no neck or tubercle. Their pointed anterior ends are tipped with cartilage. The eleventh rib has a slight angle and shallow costal groove. The twelfth rib has neither, is much shorter and slopes cranially at its vertebral end. The internal surfaces of both ribs face slightly upwards, more so in the twelfth.

Numerous muscles and ligaments are attached to the twelfth rib (**Fig. 53.10**). Quadratus lumborum and its anterior covering layer of thoracolumbar fascia are attached to the lower part of its anterior surface in its medial one-half to two-thirds; the upper part is related to the costodiaphragmatic pleural recess. The internal intercostal muscle (medially) and the diaphragm (laterally) are attached at or near the upper border. The lower border gives attachment to the middle lamella of the thoracolumbar fascia and, lateral to quadratus lumborum, to the lateral arcuate ligament and posterior lamella of the thoracolumbar fascia. The lumbocostal ligament is attached posteriorly, close to the head, connecting it to the first lumbar transverse process. The lowest levator costae, longissimus thoracis and iliocostalis are attached to the medial half of the external surface, and serratus posterior inferior, latissimus dorsi and external oblique are attached to its lateral half. The external intercostal muscle is attached along the upper border. These attachments vary: those of the internal intercostal, levator costae and erector spinae merge and those of latissimus dorsi, diaphragm and external oblique may reach the costal cartilage. The lower limit of the pleural sac crosses in front of the rib, approximately at the point where it is crossed by the lateral border of iliocostalis. Its lateral end is usually below the line of costodiaphragmatic pleural reflection and is therefore not covered by pleura.

Vascular supply and innervation The tenth and eleventh ribs are supplied by the posterior intercostal artery and branches from the musculophrenic artery. The twelfth rib is supplied by the subcostal artery. Venous drainage is via the posterior intercostal and subcostal veins, which in turn drain into the azygos system. There is additional drainage via the anterior intercostal veins (branches of the musculophrenic vein). The tenth and eleventh ribs are innervated by the corresponding intercostal nerve, and the twelfth rib is innervated by the subcostal nerve.

Ossification The tenth rib ossifies from a primary centre in the shaft and secondary centres for the head and articular parts of the tubercle. The eleventh and twelfth ribs, without tubercles, have two centres each.

Costal cartilages

Costal cartilages are the persistent, ossified anterior parts of the cartilaginous models in which the ribs develop. They are flat bars of hyaline cartilage that extend from the anterior ends of the ribs, and contribute greatly to thoracic mobility and elasticity (see Fig. 53.8). The upper seven pairs join the sternum; the eighth to tenth articulate with the lower border of the cartilage above; and the lowest two have free, pointed ends in the abdominal wall. They increase in length from the first to the seventh, and then decrease to the twelfth. They diminish in breadth from first to last, like the intercostal spaces. The costal cartilages are broad at their costal continuity and taper as they pass forwards. The first and second are of even breadth and the sixth to eighth enlarge where their margins are in contact. The first descends a little, the second is horizontal and the third ascends slightly; the others are angulated and incline up towards the sternum or cartilage above, a little anterior to their ribs.

Each costal cartilage has two surfaces, borders and ends. The anterior surface is convex, facing anterosuperiorly. The sternoclavicular articular disc, costoclavicular ligament and subclavius are attached to the first costal cartilage. Pectoralis major is attached to the medial aspect of the first six cartilages and the others are covered by the partial attachments of the anterior abdominal muscles. The posterior surface is concave and, really, posteroinferior. Sternothyroid is attached to the first cartilage, transversus thoracis is attached to the second to sixth, and transversus

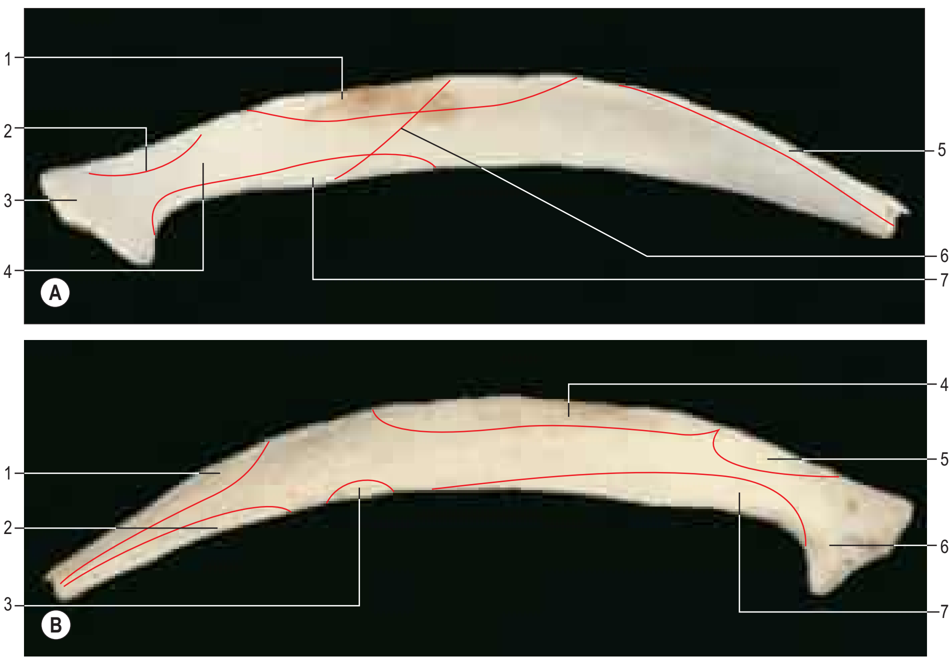


Fig. 53.10 The twelfth rib, left side. **A,** Anterior aspect. Key: 1, attachment of internal intercostal muscle; 2, attachment of costotransverse ligament; 3, head; 4, area covered by pleura; 5, attachment of diaphragm; 6, line of pleural reflection; 7, attachment of quadratus lumborum. **B,** Posterior aspect. Key: 1, attachment of latissimus dorsi; 2, attachment of external oblique; 3, attachment of serratus posterior inferior; 4, attachment of external intercostal muscle; 5, attachment of levator costae; 6, head; 7, attachment of erector spinae.

abdominis is attached to the lower six. The internal intercostal muscles and external intercostal membranes are attached to the concave superior and convex inferior borders. The inferior borders of the fifth (sometimes) and sixth to ninth cartilages project at points of greatest convexity. Oblong facets on these projections articulate with facets on slight projections from the superior borders of subjacent cartilages. The lateral end of each cartilage is continuous with its rib. The medial end of the first is continuous with the sternum; those of the six succeeding cartilages are round and articulate with shallow costal notches on the lateral margins of the sternum; those of the eighth to tenth are pointed, each connected with the cartilage above; and those of the eleventh and twelfth are pointed and free. With the exception of the fibrous synarthrosis between the first rib and sternum, all these articulations are synovial.

Costal cartilage ossification – forensic radiology

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Rib fractures

Elastic recoil of the ribs that suspend the sternum may explain the rarity of sternal fractures. Despite their pliability, the ribs are much more frequently broken, the middle ribs being the most vulnerable. Because traumatic stress is often the result of compression of the thorax, the usual site of fracture is just in front of the angle, which is the weakest point of the rib. Direct impact may fracture a rib at any point; the ends of the broken bone may be driven inwards and potentially may injure thoracic or upper abdominal viscera.

JOINTS

MANUBRIOSTERNAL JOINT

The manubriosternal joint lies between the manubrium and sternal body; it is usually a symphysis but it may be synovial, synchondrotic or synostotic. It lies approximately 7 cm below the upper border of the manubrium. The bony surfaces are covered by hyaline cartilage and connected by a fibrocartilage, which may ossify in the aged. Sometimes, the central part of the disc is absorbed and the joint appears synovial. The manubriosternal joint is connected by a fibrous membrane enveloping the entire bone. In occasional individuals over the age of 30 years, the manubrium is joined to the sternal body by bone but the intervening cartilage may be only superficially ossified; ossification becomes complete only in the aged. Early synostosis has been attributed to a persistent synchondrosis in place of a symphysis. In the newborn, union is by collagenous and elastic fibres, without chondrocytes.

Movements There is a small range of angulation between the longitudinal axes of the manubrium and body of the sternum, and limited anteroposterior displacement. The powerful ligamentous attachments in this region mean that dislocation of the manubriosternal joint is rare, although it may be associated with high-energy trauma. Dislocation is most common when the joint is synovial, whereas synchondral and synosteal types typically fracture through the manubrium.

XIPHISTERNAL JOINT

The joint between the xiphoid process and the body of the sternum process is a symphysis. It is usually transformed to a synostosis by the fortieth year but sometimes remains unchanged, even in old age.

STERNOCLAVICULAR JOINT

The sternoclavicular joint is described in Chapter 48.

COSTOVERTEBRAL, STERNOCOSTAL AND INTERCHONDRAL JOINTS

The heads of the ribs articulate with vertebral bodies (costocorporeal joints); their necks and tubercles articulate with transverse processes (costotransverse joints).

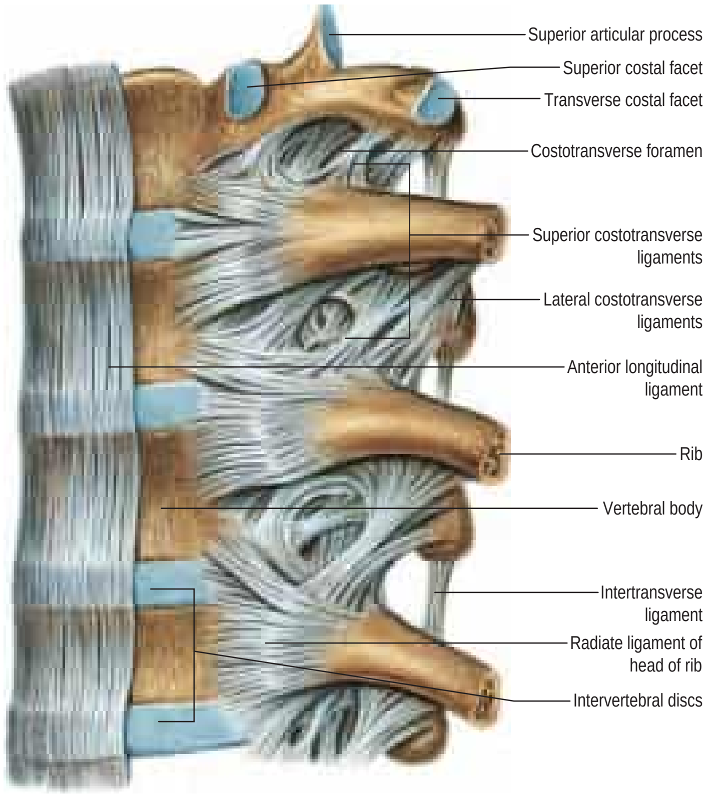

Fig. 53.11 Ligaments of the vertebral column and costovertebral joints, left lateral aspect. The lateral parts of the anterior longitudinal ligament have been removed. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer, 2013.)

Joints of costal heads

Heads of typical ribs articulate with facets (often termed demifacets) on the margins of adjacent thoracic vertebral bodies and with the intervertebral discs between them (Fig. 53.11). The first and tenth to twelfth ribs articulate with a single vertebra by a simple synovial joint. In the others, an intra-articular ligament bisects the joint, producing a double synovial compartment, so the joint is classified as both compound and complex. Often inaccurately described as plane, their articular surfaces are slightly ovoid and the upper and lower synovial articulations are obtusely angled to each other. The ligaments are capsular, radiate and intra-articular.

Fibrous capsules The fibrous capsule connects the costal head to the circumference of the articular surface formed by an intervertebral disc and the demifacets of two adjacent vertebrae. Some of the upper fibres traverse their intervertebral foramina to blend with the posterior aspects of the intervertebral discs. The posterior fibres are continuous with the costotransverse ligaments.

Radiate ligaments Radiate ligaments connect the anterior parts of each costal head to the bodies of two vertebrae and their intervening intervertebral disc. Each is attached to the head just beyond its articular surface. Superior fibres ascend to the vertebral body above, inferior to the body below. Intermediate fibres, shortest and least distinct, are horizontal and attached to the disc. The radiate ligament associated with the first rib is attached to the seventh cervical and first thoracic vertebrae. In the joints of the tenth to twelfth ribs, which articulate with single vertebrae, the radiate ligament is attached to the numbered vertebra and the one above.

Intra-articular ligament The intra-articular ligament is a short flat band attached laterally to the crest between the costal articular facets and medially to the intervertebral disc, dividing the joint. The ligament is absent from the first and tenth to twelfth joints.

Costotransverse joints

The facet of a costal tubercle articulates reciprocally with the transverse process of its corresponding vertebra (Fig. 53.12). The eleventh and twelfth ribs lack this articulation. In the upper five or six joints, articular

When present, marginal and central subperichondrial ossification patterns seen on X-ray afford a relatively simple, rapid, inexpensive and accurate means of identifying sex in completely or partially skeletonized and unidentified human material, providing the subject is over 20 years of age. The predictive ability of this technique lies in a range of 80–90%. In males, the predictive ability of the marginal subperichondrial ossification pattern is 75–80%. In females, the predictive ability for a central subperichondrial ossification pattern is greater than 90%.

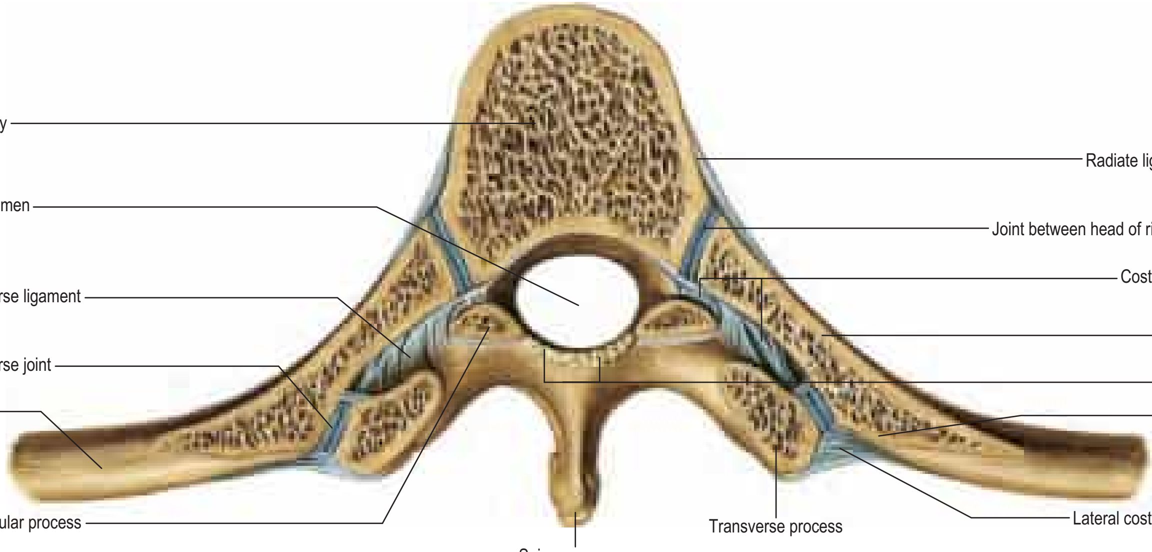

Fig. 53.12 Costovertebral joints, superior aspect.

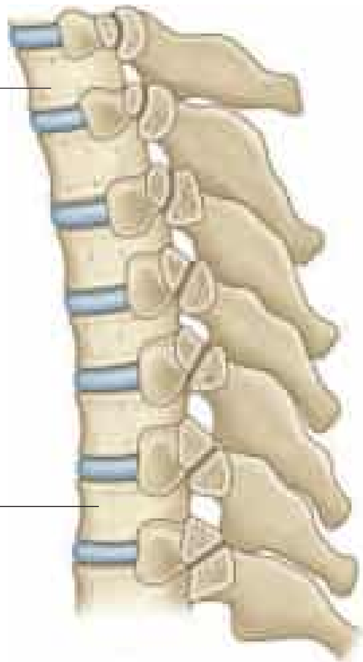

Fig. 53.13 A section through the third to the ninth costotransverse joints. Contrast the concave facets on the upper transverse processes with the less curved facets on the lower transverse processes.

surfaces are reciprocally curved but below this they are flatter (Fig. 53.13). Their ligaments are capsular, costotransverse, superior and lateral costotransverse, and accessory.

Fibrous capsule The fibrous capsule is thin and attached to articular peripheries; it has a synovial lining.

Costotransverse ligament The costotransverse ligament fills the costotransverse foramen between the neck of the rib and its adjacent corresponding transverse process. Its numerous short fibres extend back from the posterior rough surface on the neck to the anterior surface of the transverse process. A costotransverse ligament is rudimentary or absent in the eleventh and twelfth ribs.

Superior costotransverse ligament The superior costotransverse ligament has anterior and posterior layers. The anterior layer is attached between the crest of the costal neck and lower aspect of the transverse process above, and blends laterally with the internal intercostal membrane; it is crossed by the intercostal vessels and nerve. The posterior layer is attached posteriorly on the costal neck, ascending posteromedially to the transverse process above, and blends laterally with the external intercostal muscle. The first rib has no such ligament. The shaft of the twelfth rib, near its head, is connected to the base of the first lumbar transverse process by a lumbocostal ligament in series with the superior costotransverse ligaments.

Accessory ligament An accessory ligament is usually present. It lies medial to the superior costotransverse ligament, and is separated from it by the dorsal ramus of a thoracic spinal nerve and accompanying vessels. These bands are variable in their attachments, but usually pass from a depression medial to a costal tubercle to the inferior articular process immediately above. Some fibres also pass to the base of the transverse process.

Lateral costotransverse ligament The lateral costotransverse ligament is short, thick and strong, and passes obliquely from the apex of the transverse process to the rough non-articular part of the adjacent costal tubercle. The ligaments of upper ribs ascend from their transverse processes, and are shorter and more oblique than those of the lower ribs, which descend.

Movements Costal heads are so firmly tied to vertebral bodies by radiate and intra-articular ligaments that only slight gliding can occur. Strong ligaments binding costal necks and tubercles to transverse processes also limit movements at costotransverse joints to slight gliding, guided by the shape and direction of the articular surfaces (see Fig. 53.13). The facets on the tubercles of the upper six ribs are oval and vertically convex, and fit corresponding concavities on the anterior surfaces of transverse processes; consequently, up and down movements of tubercles involve rotation of costal necks about their long axes. The facets on the seventh to tenth tubercles are almost flat and face down, medially and backwards; their opposing surfaces are on the upper aspects of transverse processes and so, when these tubercles ascend, they also move posteromedially. Both sets of joints move simultaneously and in the same directions; the costal neck therefore moves as if at a single joint in which the two articulations form its ends.

In the upper six ribs, the neck moves slightly up and down but its chief movement is one of rotation about its long axis, which means that downward rotation of its anterior aspect is associated with depression, and upward rotation with elevation, of the shaft and anterior end of the rib. In the seventh to tenth ribs, the neck ascends posteromedially or descends anterolaterally, increasing or diminishing the infrasternal angle, respectively; slight rotation accompanies these movements.

Sternocostal joints

Costal cartilages articulate with small concavities on the lateral sternal borders (chondrosternal articulations) (Fig. 53.14). Perichondrium and periosteum are continuous. The first sternocostal joint is an unusual variety of synarthrosis (fibrous) and is often inaccurately called a synchondrosis. The second to seventh costal cartilages articulate by synovial joints, although articular cavities are often absent, particularly in the lower joints. Fibrocartilage covers the articular surfaces and also unites the costal cartilages and the sternum in those joints where cavities are absent. The seventh costosternal joint may be synovial or 'symphysial'. Ligaments involved are capsular, radiate sternocostal, intra-articular and costoxiphoid.

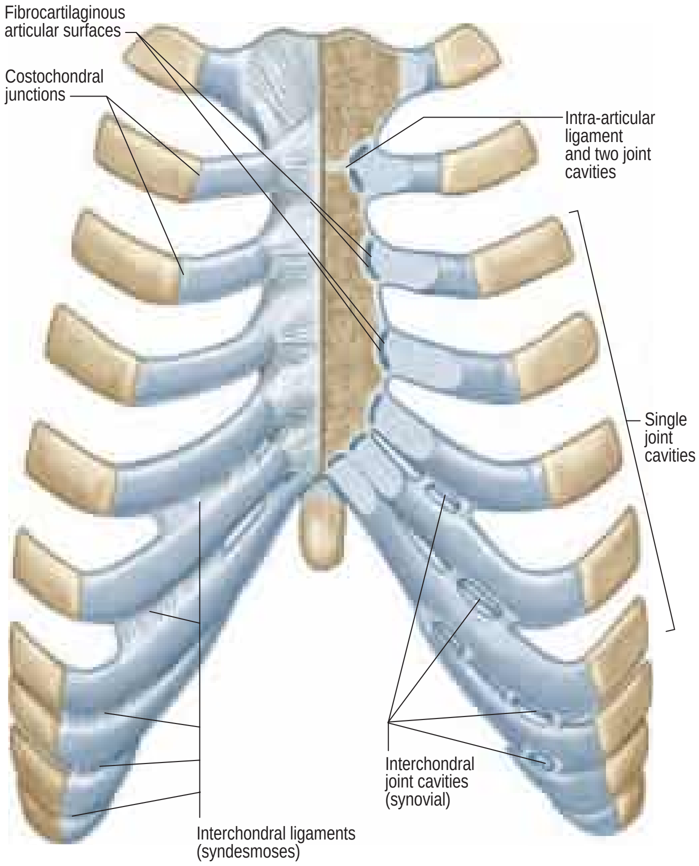

Fig. 53.14 Sternocostal and interchondral joints, anterior aspect.

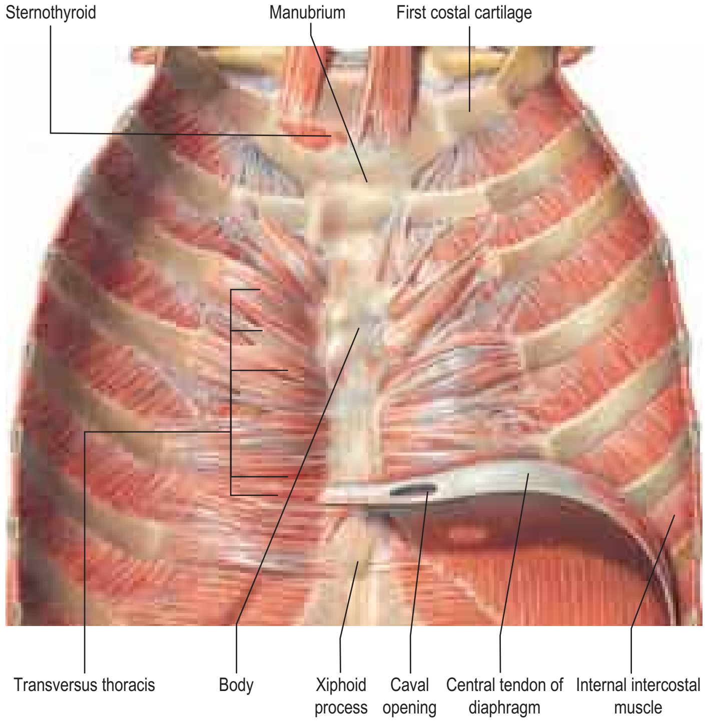

Fig. 53.15 The left transversus thoracis, exposed and viewed from its posterior aspect. The lower border of transversus thoracis is in contact with the upper border of transversus abdominis in the interval between the sternal and costal origins of the diaphragm. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer, 2013.)

Fibrous capsules Fibrous capsules surround the second to seventh sternocostal joints. They are thin, blended with the sternocostal ligaments, and strengthened above and below by fibres that connect the costal cartilages to the sternum.

Radiate sternocostal ligaments The radiate sternocostal ligaments are broad, thin bands that radiate from the anterior and posterior surfaces of the sternal ends of the costal cartilages of the true ribs to the corresponding sternal surfaces. Their superficial fibres intermingle with adjacent ligaments above and below, with those of the opposite side and with tendinous fibres of pectoralis major. Collectively, these tissues form a thick fibrous membrane around the sternum that is more marked inferiorly.

Intra-articular ligaments Intra-articular ligaments are constant only between the second costal cartilages and sternum. The ligament associated with the second costal cartilage extends from the costal cartilage to the fibrocartilage uniting the manubrium and sternal body, and is therefore intra-articular. Occasionally, the third sternal cartilage is connected with the first and second sternal segments by a similar ligament. Fibrocartilaginous strands may occur in the third and lower joints. Articular cavities may be absent at any age.

Costoxiphoid ligaments Costoxiphoid ligaments connect the anterior and posterior surfaces of the seventh (and sometimes sixth) costal cartilage to the same surfaces of the xiphoid process. They vary in length and breadth, and the posterior is less distinct.

Movements Slight gliding movements, sufficient for ventilation, occur at sternocostal joints.

Interchondral joints

Contiguous borders of the sixth to ninth costal cartilages articulate by apposition of small oblong facets. Each articulation is enclosed in a thin fibrous capsule, lined by synovial membrane with lateral and medial interchondral ligaments. Sometimes the fifth cartilage, and less commonly the ninth cartilage, articulate at their inferior borders with adjoining cartilages; this connection is usually effected by ligamentous fibres. Articulation between the ninth and tenth cartilages is never synovial and sometimes absent.

Costochondral junctions

Artificially separated from its rib, a costal cartilage has a rounded end that fits a reciprocal depression in the rib. Periosteum and perichondrium are continuous across the costochondral junctions and the collagen of the osseous and cartilaginous matrices blend. No movement occurs at costochondral junctions.

MUSCLES

The intrinsic muscles of the chest wall are the intercostals, subcostales, transversus thoracis, levatores costarum, serratus posterior and, occasionally, sternalis. Scapular muscles and muscles connecting the upper limb, chest wall and vertebrae, i.e. trapezius, latissimus dorsi, rhomboids major and minor, levator scapulae, pectorales major and minor, subclavius, supra- and infraspinatus, teres major and teres minor, are described in detail in Chapter 48.

INTRINSIC CHEST WALL MUSCLES

Intercostal muscles

The intercostal muscles are thin multiple layers of muscular and tendinous fibres that occupy the intercostal spaces; their names are derived from their spatial relationship, i.e. the external, internal and innermost intercostals (Fig. 53.15).

External intercostals

Eleven pairs of external intercostals extend from the tubercles of the ribs, where they blend with the posterior fibres of the superior costotransverse ligaments, almost to the costal cartilages, where each continues forwards to the sternum as an aponeurotic layer, the external intercostal membrane. Each muscle passes from the lower border of one rib to the upper border of the rib below; their fibres are directed obliquely downwards and laterally at the back of the thorax, and downwards, forwards and medially at the front. In the upper two or three spaces, they do not quite reach the ends of the rib, and in the lower two spaces, they extend to the free ends of the costal cartilages. The external intercostals are thicker than the internal intercostals.

SECTION 7

Innervation External intercostals are supplied by the adjacent intercostal nerves.

Action External intercostals are believed to act with the internal intercostals (Ch. 55).

Internal intercostals

Eleven pairs of internal intercostals begin anteriorly at the sternum, in the interspaces between the cartilages of the true ribs, and at the anterior extremities of the cartilages of the 'false' ribs. Their greatest thickness lies in this intercartilaginous or parasternal part. They continue back as far as the posterior costal angles, where each is replaced by an aponeurotic layer, the internal intercostal membrane, that is continuous posteriorly with the anterior fibres of a superior costotransverse ligament, and anteriorly with the fascia between the internal and external intercostal muscles. Each muscle descends from the floor of a costal groove and adjacent costal cartilage, and inserts into the upper border of the rib below; their fibres are directed obliquely, nearly at right angles to those of the external intercostal muscles.

Innervation Internal intercostals are supplied by the adjacent intercostal nerves.

Action Internal intercostals are believed to act with the external intercostals (p. 974).

Innermost intercostals

The innermost intercostals were once regarded as internal laminae of the internal intercostal muscles, and fibres in the two layers do coincide in direction. Each muscle is attached to the internal aspects of two adjoining ribs. They are insignificant, and sometimes absent, at the highest thoracic levels but become progressively more substantial below this, typically extending through the middle two quarters of the lower intercostal spaces. Posteriorly, the innermost intercostals, in those spaces where they are well developed, may come together with the corresponding subcostales. The innermost intercostals are related internally to the endothoracic fascia and parietal pleura, and externally to the intercostal nerves and vessels.

Innervation Innermost intercostals are supplied by the adjacent intercostal nerves.

Action Innermost intercostals are believed to act with the internal intercostals (Ch. 55).

Subcostales

Subcostales consist of muscular and aponeurotic fasciculi and are usually well developed only in the lower part of the thorax. Each descends from the internal surface of one rib, near its angle, to the internal surface of the second or third rib below. Their fibres run parallel to those of the internal intercostals and, like the innermost intercostals, they lie between the intercostal vessels and nerves and the pleura.

Innervation Subcostales are supplied by the adjacent intercostal nerves.

Action Subcostales depress the ribs.

Transversus thoracis

Transversus thoracis (triangularis sternae, sternocostalis) spreads over the internal surface of the anterior thoracic wall (see Fig. 53.15). It arises from the lower third of the posterior surface of the sternum, the xiphoid process and the costal cartilages of the lower three or four true ribs near their sternal ends. The fibres diverge and ascend laterally as slips that pass into the lower borders and inner surfaces of the costal cartilages of the second, third, fourth, fifth and sixth ribs. The lowest fibres are horizontal and are contiguous with the highest fibres of transversus abdominis; the intermediate fibres are oblique; and the highest are almost vertical. Transversus thoracis varies in its attachments, not only between individuals but even on opposite sides of the same individual. Like the innermost intercostals and subcostales, transversus thoracis separates the intercostal nerves from the pleura.

Innervation Transversus thoracis is supplied by the adjacent intercostal nerves.

Action Transversus thoracis draws down the costal cartilages to which it is attached.

Levatores costarum

Levatores costarum are strong bundles, 12 on each side, which arise from the tips of the transverse processes of the seventh cervical and first to eleventh thoracic vertebrae. They pass obliquely downwards and laterally, parallel with the posterior borders of the external intercostals. Each is attached to the upper edge and external surface of the rib immediately below the vertebra from which it takes origin, between the tubercle and the angle (levatores costarum breves). Each of the four lower muscles divides into two fasciculi; one is attached as already described, and the other descends to the second rib below its origin (levatores costarum longi).

Innervation Levatores costarum are supplied by the lateral branches of the dorsal rami of the corresponding thoracic spinal nerves.

Action Levatores costarum elevate the ribs but their importance in ventilation is disputed. They are also said to act from their costal attachments as rotators and lateral flexors of the vertebral column.

Serratus posterior superior

Serratus posterior superior (Fig. 53.16) is a thin quadrilateral muscle, external to the upper posterior part of the thorax. It arises by a thin aponeurosis from the lower part of the nuchal ligament, the spines of the seventh cervical and upper two or three thoracic vertebrae, and their supraspinous ligaments. It descends laterally and ends in four digitations attached to the upper borders and external surfaces of the second, third, fourth and fifth ribs, just lateral to their angles. It is superficial to the thoracic part of the thoracolumbar fascia and deep to the rhomboids. The number of digitations can vary from three to six, and the muscle may even be absent.

Innervation Serratus posterior superior is innervated by the second, third, fourth and fifth intercostal nerves.

Action The attachments of serratus posterior superior clearly indicate that it could elevate the ribs; its role in humans is uncertain.

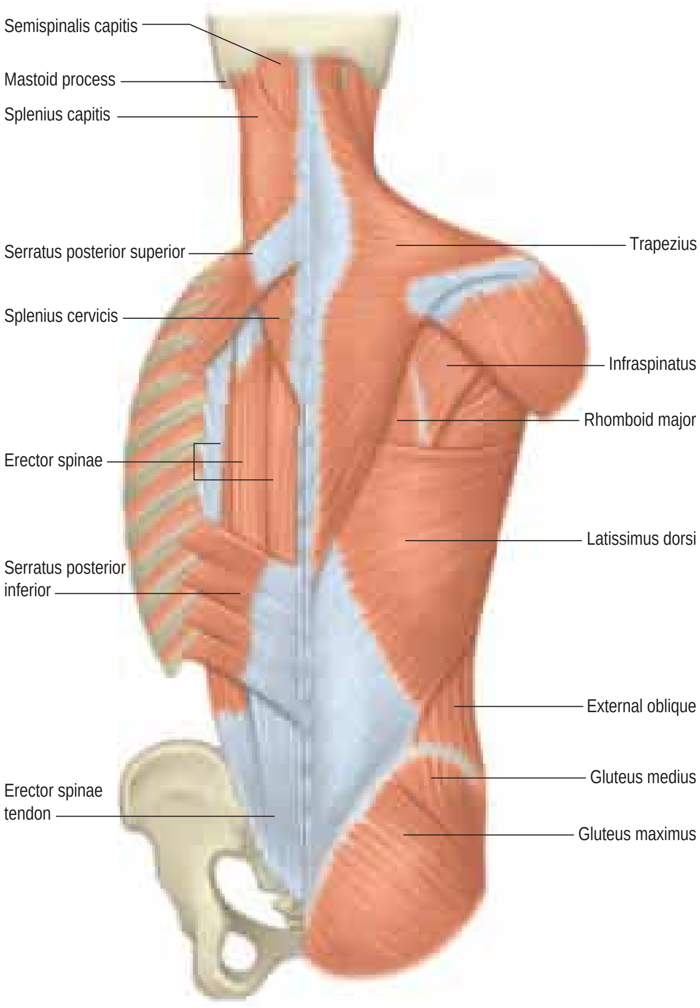

Fig. 53.16 Superficial (extrinsic) muscles of the back.

Serratus posterior inferior

Serratus posterior inferior (Fig. 53.16) is a thin, irregularly quadrilateral muscle at the junction of the thoracic and lumbar regions. It arises from the spines of the lower two thoracic and upper two or three lumbar vertebrae and their supraspinous ligaments by a thin aponeurosis that blends with the lumbar part of the thoracolumbar fascia. It ascends laterally and its four digitations pass into the inferior borders and outer surfaces of the lower four ribs, a little lateral to their angles. There may be fewer digitations and, in rare cases, the entire muscle may be absent.

Innervation Serratus posterior inferior is innervated by ventral rami of the ninth, tenth, eleventh and twelfth thoracic spinal nerves.

Action Serratus posterior inferior draws the lower ribs downwards and backwards, although possibly not in ventilation.

Sternalis

Sternalis is an anatomical variation well known to anatomists but relatively unknown to clinicians and surgeons (Snosek et al 2014). The muscle appears as a parasternal mass deep to the superficial fascia of the anterior thoracic wall and superficial to the pectoral fascia overlying pectoralis major. It may be a cord-like, flat band or irregular and flame-like in shape, is almost twice as commonly unilateral, and occurs more often on the right side. Various attachment sites have been described in an extensive anatomical literature, including the sternum, inferior border of the clavicle, sternocleidomastoid fascia, pectoralis major, and the upper ribs and their costal cartilages, all superiorly, and the lower ribs and their costal cartilages, pectoralis major, the rectus sheath and the external abdominal oblique aponeurosis, all inferiorly. The muscle occurs occasionally in the general population, but there is great variation both within and between different geographic populations. Its superficial location makes it an ideal candidate for utilization as a muscular flap in plastic reconstruction of the head and neck region.

The aetiology is unknown; sternalis may be an example of a much larger group of variations, including pectoralis minimus, pectoralis tertius, infraclavicularis and chondroepitrochlearis, that are thought to be caused by a disturbance of the normal processes of pectoral muscle development.

The relationship of sternalis to pectoralis major may cause a diagnostic dilemma during breast surgery, mammography, computed tomography and magnetic resonance imaging scans because its appearance mimics tumour pathology of the region.

Mechanism of thoracic cage movement

Breathing involves changing the thoracic volume by altering the vertical, transverse and anteroposterior dimensions of the thorax (p. 974). The diaphragm is the key muscle in this process. Its muscle fibres descend from their relatively 'high' anterior sternocostal attachments steeply to the central tendon and obliquely to their complex 'low' posterior attachments (see Fig. 55.1). The central tendon is fixed: when the diaphragm contracts, it allows the lower ribcage to move inferiorly and anteriorly without any change to the curvature of the diaphragm. The intercostal muscles maintain the rigidity of the chest wall. The external and internal intercostals, transversus thoracis, subcostales, levatores costarum, serratus posterior superior and serratus posterior inferior can elevate or depress the ribs, and hence can act as accessory muscles of ventilation.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE OF THE CHEST WALL

ARTERIES

Muscles of the thoracic wall receive their blood supply from the internal thoracic artery, either directly or via the musculophrenic artery, the superior intercostal artery, descending thoracic aorta, and the subcostal and superior thoracic arteries. Additional contributions come from vessels that supply the proximal muscles of the upper limb: namely, the suprascapular, superficial cervical, thoraco-acromial, lateral thoracic and subscapular arteries.

Internal thoracic artery

The internal thoracic artery (internal mammary artery) arises inferiorly from the first part of the subclavian artery, approximately 2 cm above the sternal end of the clavicle and opposite the root of the thyrocervical trunk (see Fig. 51.2). It descends behind the first six costal cartilages, 1 cm from the lateral sternal border and divides at the level of the sixth intercostal space into musculophrenic and superior epigastric branches. The superior epigastric artery is described in Chapter 61.

Relations At first, the internal thoracic artery descends anteromedially behind the sternal end of the clavicle, the internal jugular and brachiocephalic veins, and the first costal cartilage. As it enters the thorax, the phrenic nerve crosses it obliquely from its lateral side, usually in front. The artery then descends almost vertically to its bifurcation, lying behind pectoralis major, the first six costal cartilages, external intercostal membranes, internal intercostals and terminations of the upper six intercostal nerves. It is separated from the pleura, down to the second or third cartilage, by a strong layer of fascia, and below this by transversus thoracis. The artery is accompanied by a chain of lymph nodes and by venae comitantes that unite at about the third costal cartilage into a single vein medial to the artery. Its intermediate branches are sternal, anterior intercostal and perforating.

Sternal branches Sternal branches are distributed to transversus thoracis, the periosteum of the posterior sternal surface and the sternal red bone marrow. These branches, together with small branches of the pericardiacophrenic artery, anastomose with branches of the posterior intercostal and bronchial arteries to form a subpleural mediastinal plexus.

Anterior intercostal branches Anterior intercostal arteries are distributed to the upper six intercostal spaces. They pass laterally along the borders of the space and anastomose with the posterior intercostal arteries and their collateral branches (see Fig. 53.19). The anterior intercostals usually arise from the internal thoracic artery as single vessels that promptly divide into two branches, one passing superiorly, the other inferiorly within each intercostal space. Occasionally, one branch passes to the space above and one to the space below. (The anterior intercostals may sometimes arise as two separate branches from the internal thoracic artery.) The arteries lie at first between the pleura and the internal intercostals, then between the innermost and the internal intercostals (**Fig. 53.18**). They supply the intercostal muscles and send branches through them to the pectoral muscles, breast and skin.

Perforating branches Perforating branches traverse the upper five or six intercostal spaces with the anterior cutaneous branches of the corresponding intercostal nerves. They pierce and supply pectoralis major, and then curve laterally to become direct cutaneous vessels that supply the skin. These cutaneous vessels provide the anatomical basis for surgically raising the deltopectoral skin flap that is used for reconstructing areas of missing tissue in the head and neck. The second to fourth branches supply the breast and become enlarged during lactation.

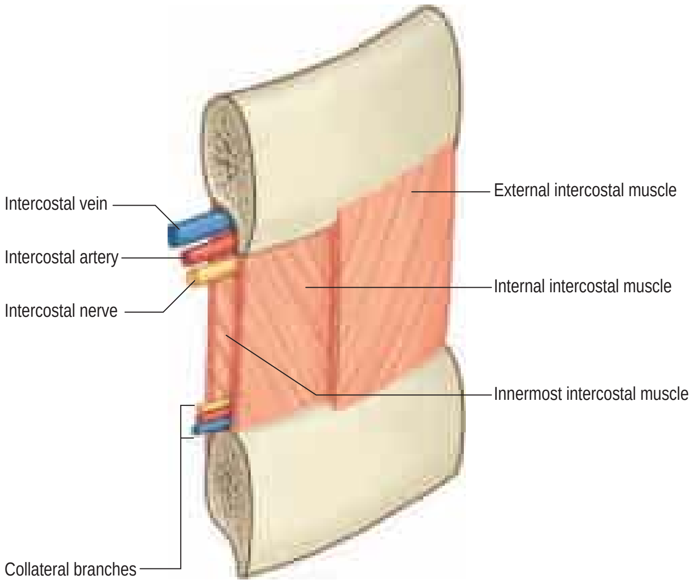

Fig. 53.18 Dissection of part of an intercostal space, external aspect, showing the position of the intercostal vessels and nerve and their collateral branches relative to the intercostal muscles.

The mammographic appearance of sternalis muscle is variable. Typically visible in the medial aspect of the breast on a craniocaudal (CC) mammogram, it appears as a small soft tissue density/mass abutting the chest wall. Its margins and shape are variable. Sternalis is not usually seen on standard mediolateral oblique or mediolateral views.

Musculophrenic artery The musculophrenic artery passes inferolaterally behind the seventh to ninth costal cartilages, traverses the diaphragm near the ninth, and ends near the last intercostal space. It anastomoses with the inferior phrenic and lower two posterior intercostal arteries and ascending branches of the deep circumflex iliac arteries. Two anterior intercostal arteries branch from it for each of the seventh to ninth intercostal spaces, and are distributed similarly to their counterparts in the higher spaces. The musculophrenic artery also supplies the lower part of the pericardium and the abdominal muscles.

Clinical significance of the internal thoracic artery

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Superior intercostal artery

The superior intercostal artery arises from the costocervical trunk. It descends between the pleura and the necks of the first and second ribs, and anastomoses with the third posterior intercostal artery. Crossing the neck of the first rib, it lies medial to the ventral branch of the first thoracic spinal nerve, which it crosses at a lower level, and lateral to the stellate ganglion. In the first space, it gives off the first posterior intercostal artery, which has a similar distribution to the lower posterior intercostal arteries. It descends to become the second posterior intercostal artery, usually joining a branch from the third. The artery is not constant, and is more common on the right; when absent, it is replaced by a direct branch from the aorta.

Posterior intercostal arteries

There are usually nine pairs of posterior intercostal arteries. They arise from the posterior aspect of the descending thoracic aorta and are distributed to the lower nine intercostal spaces. Right posterior intercostal arteries are longer because the aorta deviates to the left (Fig. 53.19); they cross the vertebral bodies behind the oesophagus, thoracic duct and azygos vein, right lung and pleura. Left posterior intercostal arteries turn backwards on the vertebral bodies in contact with the left lung and pleura; the upper two are crossed by the left superior intercostal vein, and the lower by the hemiazygos and accessory hemiazygos veins. The further course of the arteries is the same on both sides. The sympathetic trunk lies anterior to all of the arteries, and the splanchnic nerves descend in front of the lower arteries.

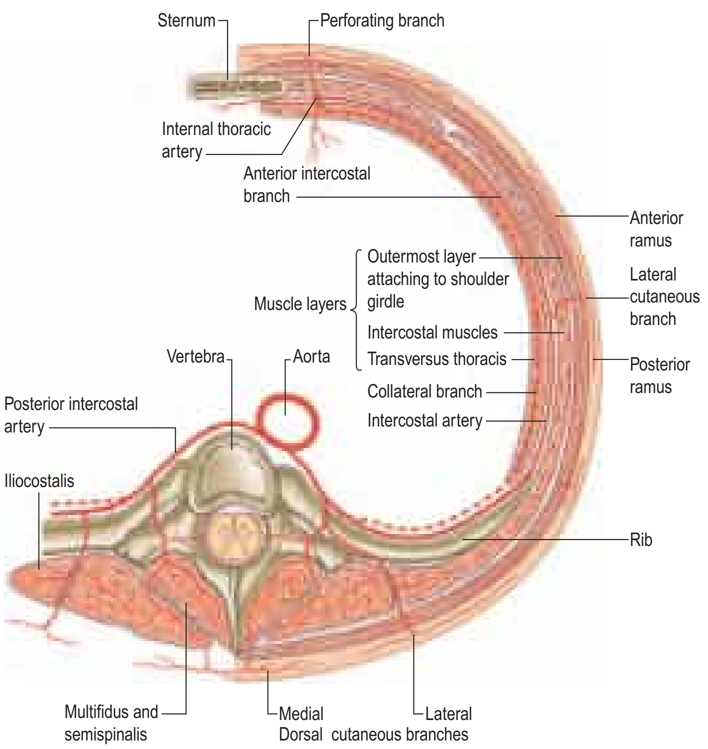

Fig. 53.19 A schematic transverse section through a lower intercostal space to show the branches of a typical intercostal artery. The usual branching arrangement of the dorsal cutaneous branch is shown to the right of the spine and a common variant is shown to the left. (With permission from Cormack GC, Lamberty BGH 1994 The Arterial Anatomy of Skin Flaps, 2nd ed. Edinburgh: Churchill Livingstone.)

Each artery crosses its intercostal space obliquely towards the angle of the rib above and continues forwards in its costal groove. At first between the pleura and internal intercostal membrane as far as the costal angle, it passes between the internal intercostal and innermost intercostal muscles (see Fig. 53.18), anastomosing with an anterior intercostal branch from either the internal thoracic or musculophrenic artery. Each artery has a vein above and a nerve below, except in the upper spaces, where the nerve at first lies above the artery. The third posterior intercostal artery anastomoses with the superior intercostal artery and may provide the major supply to the second space. The lower two arteries continue anteriorly into the abdominal wall, where they anastomose with the subcostal, superior epigastric and lumbar arteries. Each posterior intercostal artery has dorsal, collateral, muscular and cutaneous branches.

Dorsal branch Each dorsal branch runs dorsally between the necks of adjoining ribs; a vertebral body and superior costotransverse ligament lie medial and lateral, respectively. Each dorsal branch gives off a spinal branch that enters the vertebral canal via the intervertebral foramen and supplies the vertebra, spinal cord and meninges, anastomosing with the spinal arteries above and below and with its contralateral fellow. It then divides into a medial and a lateral dorsal musculocutaneous branch (occasionally, these arise separately from the posterior intercostal artery rather than from a common trunk). The medial branch crosses a transverse process with the medial dorsal branch of a thoracic spinal nerve to supply spinalis, longissimus thoracis and an area of overlying skin. The lateral branch supplies longissimus thoracis and iliocostalis, and the medial aspects of latissimus dorsi and trapezius, in addition to an area of overlying skin.

Collateral intercostal branch A collateral intercostal branch arises near the costal angle and descends to the upper border of the subjacent rib, along which it courses to anastomose with an anterior intercostal branch of the internal thoracic or musculophrenic artery.

Muscular branches Muscular branches supply the intercostal and pectoral muscles and serratus anterior, anastomosing with the superior and lateral thoracic branches of the axillary artery. Lateral cutaneous branches accompany the same branches of the thoracic spinal nerves. Mammary branches from the vessels in the second to fourth spaces supply the pectoral muscles, breast tissue and skin; they enlarge during lactation.

Lateral cutaneous branch The lateral cutaneous branch is given off in the posterior part of the intercostal space and travels anteriorly for a few centimetres with an accompanying vein and the lateral cutaneous branch of the equivalent intercostal nerve. This neurovascular bundle pierces the intercostal muscles and emerges lower down between the interdigitations of serratus anterior and external oblique. It divides into anterior and posterior rami, which contribute to the blood supply of the skin of the lateral trunk; these vessels are less significant in the upper three intercostal spaces.

Unnamed branches Other unnamed branches supply tissues of the thoracic wall, e.g. costal periosteum, bone and bone marrow of the ribs, tissues of synovial and synarthrodial joints, and the parietal pleura.

Clinical significance of the posterior intercostal arteries

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VEINS

Internal thoracic veins

The internal thoracic veins are venae comitantes of the inferior half of the internal thoracic artery. Near the third costal cartilages, the veins unite and ascend medial to the artery to end in their appropriate brachiocephalic vein. Tributaries correspond to branches of the artery and include a pericardiacophrenic vein. The internal thoracic veins have several valves.

Left superior intercostal vein

The left superior intercostal vein drains the second and third (sometimes fourth) left posterior intercostal veins. It ascends obliquely forwards across the left aspect of the aortic arch, lateral to the left vagus

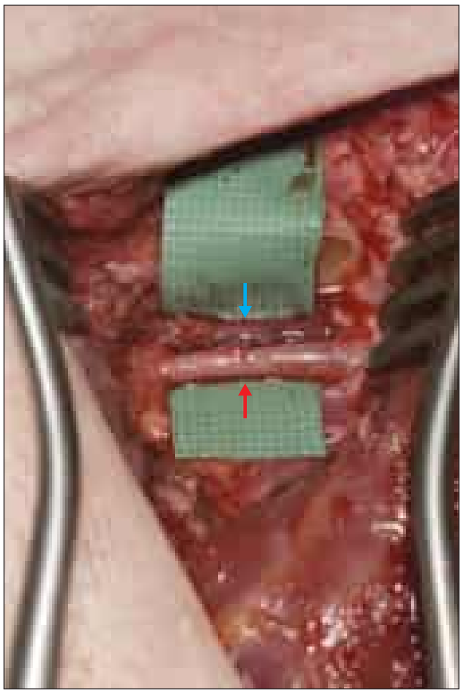

Fig. 53.17 The internal thoracic vessels prepared for micro-anastomosis beneath the third costal cartilage. The dissected internal thoracic artery (red arrow) and vein (blue arrow) are shown after removal of the right third costal cartilage. These vessels are divided to provide recipient blood supply for a free tissue transfer.

The internal thoracic artery has a number of clinical uses but perhaps most significantly is the vessel of choice for coronary artery bypass grafting (Fig. 53.17). It has become a popular recipient vessel for free flap transfers to the chest wall, principally for breast reconstruction (Ninković et al 1995), and has largely superseded the thoracodorsal vessels for this purpose. It is routinely accessed by removing a portion of the third costal cartilage, although some surgeons prefer to dissect the vessels between the ribs, with a rib-sparing approach (Parrett et al 2008, Malata et al 2011) or even utilizing the anterior perforating branches as they pierce pectoralis major (Park et al 2003, Munhoz et al 2004). The anterior perforating branches form the vascular pedicles for fasciocutaneous flaps raised from the chest wall that have been used for many decades in head and neck reconstruction (Bakamjian et al 1971, Neligan et al 2007). The superior epigastric artery is the main vascular pedicle for the superiorly based rectus abdominis flap, a muscle or myocutaneous flap useful in breast, sternal and chest wall reconstruction (Hartrampf et al 1982). During CT-guided lung and mediastinal biopsies, the internal thoracic vessels must be avoided: exsanguination has been reported with Trucut needle trauma. The approach must be at least 2 cm lateral to the sternal margin. The vessels are not seen easily on non-contrast fluoroscopic CT images.

Catheter embolization is an interventional radiology technique that can be used to block branches of the posterior intercostal arteries when they are bleeding, e.g. secondary to blunt trauma usually associated with rib fractures, or when there is an abnormality of the vertebrae supplied by the dorsal branch of the posterior intercostal artery. In such cases, the posterior intercostal artery may be accessed by placing a catheter in the thoracic aorta, usually by puncturing the femoral artery in the groin. A smaller catheter (referred to as a superselective catheter) may then be placed in the posterior intercostal artery and embolic material (which can be metallic coils or glue) injected to block the artery. This can cause cessation of bleeding in cases of trauma and provide pain relief when the vertebral body is infiltrated with tumour (Fig. 53.20).

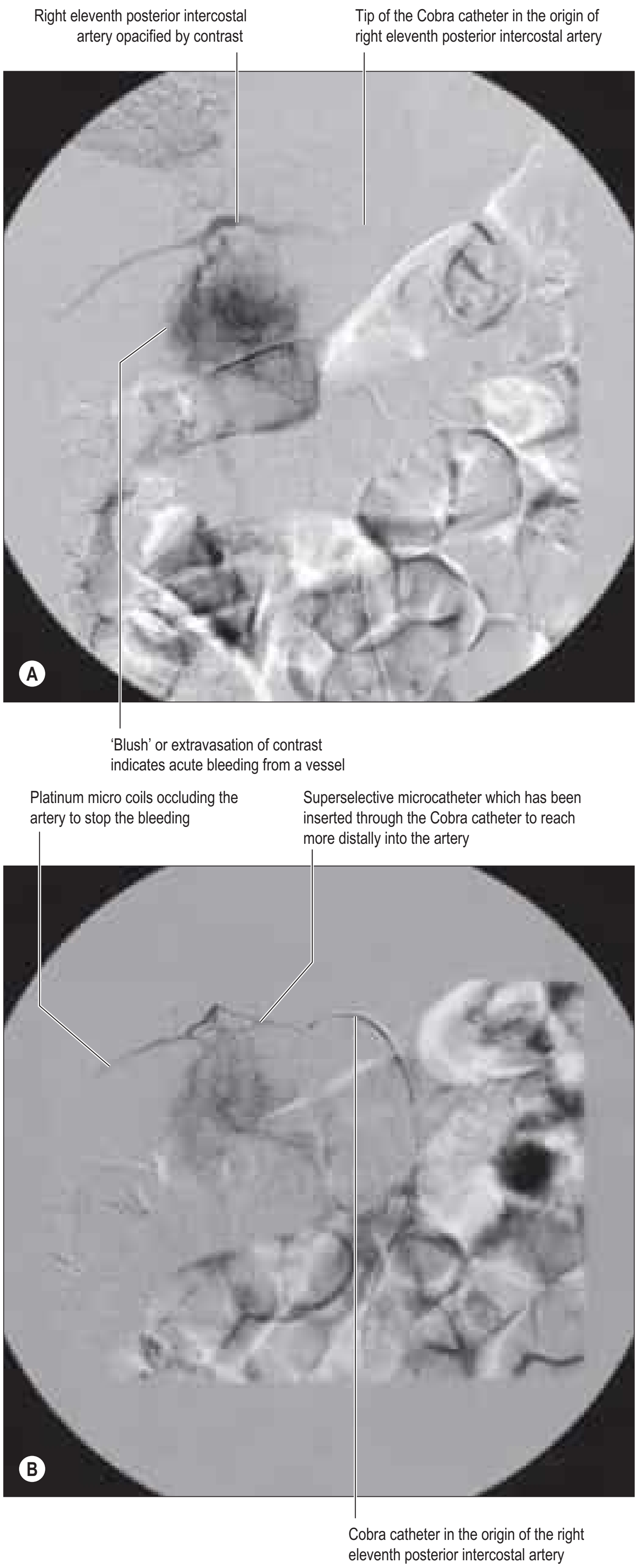

Fig. 53.20 A,B, Pre- and post-embolization images after selective catheterization of the right eleventh posterior intercostal artery.

CHAPTER 53

and medial to the left phrenic nerve, to open into the left brachiocephalic vein. It usually receives the left bronchial veins, sometimes the left pericardiacophrenic vein, and connects inferiorly with the accessory hemiazygos vein. It may be seen on a chest X-ray as the 'aortic nipple', not to be confused with a small lung nodule.

Posterior intercostal veins

The posterior intercostal veins accompany their arteries in 11 pairs. Approaching the vertebral column, each vein receives a posterior tributary that returns blood from the dorsal muscles and skin and the vertebral venous plexuses. On both sides, the first posterior intercostal vein ascends anterior to the neck of the first rib, arching forwards above the pleural dome to end in the ipsilateral brachiocephalic or vertebral vein. On the right side, the second, third and often the fourth veins form a right superior intercostal vein that joins the arch of the azygos vein. The lower spaces drain directly into the azygos vein. On the left side, the second, third and sometimes the fourth veins form a left superior intercostal vein. Veins from the fourth or fifth to eighth intercostal spaces end in the accessory hemiazygos vein, and veins from the ninth to the eleventh intercostal spaces drain into the hemiazygos vein.

Posterior intercostal veins are so called to distinguish them from small anterior intercostal veins, which are tributaries of the internal thoracic and musculophrenic veins.

LYMPHATIC DRAINAGE

Superficial lymphatic vessels of the thoracic wall ramify subcutaneously and converge on the axillary nodes (see Fig. 53.22). Those superficial to trapezius and latissimus dorsi unite to form 10 or 12 trunks, which end in the subscapular nodes. Those in the pectoral region, including vessels from the skin covering the periphery of the breast and its subareolar plexus, run back, collecting those superficial to serratus anterior, to reach the pectoral nodes. Vessels near the lateral sternal margin pass between the costal cartilages to the parasternal nodes and anastomose across the sternum, providing a route for contralateral nodal spread in medially located breast carcinoma. A few vessels from the upper pectoral region ascend over the clavicle to the inferior deep cervical nodes. Lymph from the deeper tissues of the thoracic walls drains mainly to the parasternal, intercostal or diaphragmatic nodes.

Parasternal (internal thoracic) nodes

There are four or five parasternal nodes along each internal thoracic artery at the anterior ends of the intercostal spaces. They drain afferents from the breast, deeper structures of the supra-umbilical anterior abdominal wall, the superior hepatic surface (through a small group of nodes behind the xiphoid process) and deeper parts of the anterior thoracic wall. Their efferents usually unite with those from the tracheobronchial and brachiocephalic nodes to form the bronchomediastinal trunk. The latter may open on either side, directly into the jugulosubclavian junction, into either great vein near the junction, the right subclavian trunk or lymphatic duct, or the thoracic duct on the left.

Intercostal nodes

Intercostal nodes occupy the intercostal spaces near the heads and necks of the ribs. They receive deep lymph vessels from the posterolateral aspects of the chest wall and breast, some of which are interrupted by small lateral intercostal nodes. Efferents of nodes in the lower 4–7 spaces unite into a trunk that descends to the abdominal confluence of lymph trunks or to the start of the thoracic duct. Efferents of nodes in the left upper spaces end in the thoracic duct; those of the right upper spaces end in one of the right lymph trunks.

Diaphragmatic nodes

Located on the thoracic surface of the diaphragm, these nodes are arranged in anterior, right and left lateral, and posterior groups.

Anterior group The anterior group consists of two or three small nodes behind the base of the xiphoid process, draining the convex hepatic surface, and one or two nodes on each side near the junction of the seventh rib and cartilage, which receive anterior lymph vessels from the diaphragm. The anterior group drains to the parasternal nodes.

Lateral groups The lateral groups each contain two or three nodes, and lie close to the point where the phrenic nerves enter the diaphragm. On the right, some nodes lie within the fibrous pericardium anterior to the intrathoracic end of the inferior vena cava. Their afferents drain the central diaphragm; those on the right also drain the convex surface of the liver, and their efferents pass to the posterior mediastinal, parasternal and brachiocephalic nodes.

Posterior group The posterior group consists of a few nodes that lie on the posterior aspect of the crura and connect with the lateral aortic and posterior mediastinal nodes.

Lymphatic drainage of deeper tissues

Collecting vessels of the deeper thoracic tissues include lymphatics that drain the muscles attached to the ribs. Most end in axillary nodes but some from pectoralis major also drain to the parasternal nodes. Intercostal vessels drain the intercostal muscles and parietal pleura; those from the anterior thoracic wall and pleura end in the parasternal nodes, and their posterior counterparts drain to intercostal nodes.

Vessels from the diaphragm form two plexuses, thoracic and abdominal, which anastomose freely, especially in areas covered by pleura and peritoneum, respectively. The thoracic plexus unites with lymph vessels draining the costal and mediastinal pleura. Its efferents are anterior, draining to the anterior diaphragmatic nodes near the junctions of the seventh rib and cartilage; middle, draining to nodes on the oesophagus and around the end of the inferior vena cava; and posterior, draining to nodes around the aorta at the point where it leaves the thorax. The abdominal plexus anastomoses with the hepatic lymphatics and peripherally with those of the subperitoneal tissue. Efferents from its right half end in a group of nodes on the inferior phrenic artery, or in the right lateral aortic nodes. Those from the left half of the abdominal diaphragmatic plexus pass to the pre-aortic, lateral aortic and terminal oesophageal nodes.

INNERVATION OF THE CHEST WALL

THORACIC VENTRAL SPINAL RAMI

There are 12 pairs of thoracic ventral rami. The upper 11 lie between the ribs (intercostal nerves) and the twelfth lies below the last rib (subcostal nerve). Each is connected with the adjoining ganglion of the sympathetic trunk by grey and white rami communicantes; the grey ramus joins the nerve proximal to the point at which the white ramus leaves it (Fig. 53.21). Intercostal nerves are distributed primarily to the thoracic and abdominal walls. The first two nerves supply fibres to the

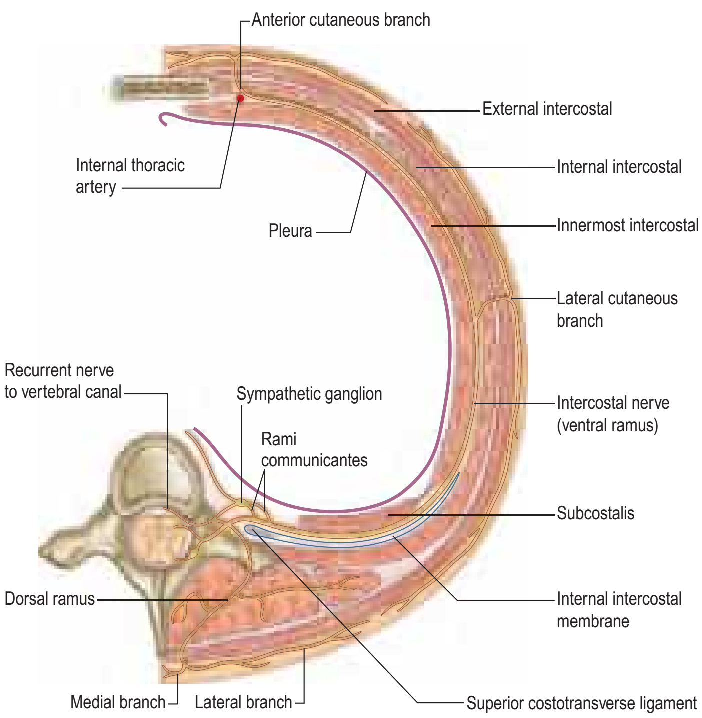

Fig. 53.21 The course of a typical intercostal nerve. The muscular and the collateral branches are not shown.

upper limb in addition to their thoracic branches, the next four supply only the thoracic wall, and the lower five supply both thoracic and abdominal walls. The subcostal nerve is distributed to the abdominal wall and the gluteal skin. Communicating branches link the intercostal nerves posteriorly in the intercostal spaces, and the lower five nerves communicate freely in the abdominal wall.

First to sixth thoracic ventral rami

The first thoracic ventral ramus divides unequally: a large branch ascends across the neck of the first rib, lateral to the superior intercostal artery, to enter the brachial plexus, and a smaller branch (the first intercostal nerve) runs in the first intercostal space and terminates as the first anterior cutaneous nerve of the thorax. A lateral cutaneous branch pierces the chest wall anterior to serratus anterior and supplies the axillary skin; it may communicate with the intercostobrachial nerve and sometimes joins the medial cutaneous nerve of the arm. The first thoracic ramus often receives a connecting ramus from the second, which ascends in front of the neck of the second rib.

The second to sixth thoracic ventral rami pass forwards in their intercostal spaces below the intercostal vessels. Posteriorly, they lie between the pleura and external intercostal membranes, but they run mainly between the internal intercostals and the subcostales/innermost intercostals. Near the sternum, they cross anterior to the internal thoracic vessels and transversus thoracis, pierce the internal intercostals, the external intercostal membranes and pectoralis major, and terminate as the anterior cutaneous nerves of the thorax. The second anterior cutaneous nerve may be connected to the medial supraclavicular nerves of the cervical plexus, and twigs from the sixth intercostal nerve supply abdominal skin in the upper part of the infrasternal angle.

Branches Numerous slender muscular filaments supply the intercostals, serratus posterior superior and transversus thoracis. Anteriorly, some cross the costal cartilages from one intercostal space to another. Each intercostal nerve gives off a collateral and a lateral cutaneous branch before it reaches the angle of the adjoining ribs (see Fig. 53.21). The collateral branch follows the inferior border of its space in the same intermuscular plane as the main nerve, which it may rejoin before it is distributed as an additional anterior cutaneous nerve. The lateral cutaneous branch accompanies the main nerve a little way and then pierces the intercostal muscles obliquely. With the exception of the lateral cutaneous branches of the first and second intercostal nerves, each divides into anterior and posterior rami that subsequently pierce serratus anterior. Anterior branches run forwards over the border of pectoralis major to supply the overlying skin; those of the fifth and sixth also supply twigs to a variable number of upper digitations of external oblique. Posterior branches run backwards and supply the skin over the scapula and latissimus dorsi.

The lateral cutaneous branch of the second intercostal nerve is the intercostobrachial nerve (see Fig. 48.34). It crosses the axilla to gain the medial side of the arm and joins a branch of the medial cutaneous nerve of the arm. It then pierces the deep fascia of the arm, and supplies the skin of the upper half of the posterior and medial parts of the arm, communicating with the posterior cutaneous branch of the radial nerve. Its size is in inverse proportion to the size of the medial cutaneous nerve. A second intercostobrachial nerve often branches off from the anterior part of the third lateral cutaneous nerve and sends filaments to the axilla and the medial side of the arm.

Seventh to eleventh thoracic ventral rami

The ventral rami of the seventh to eleventh thoracic nerves are continued anteriorly from the intercostal spaces into the abdominal wall; their further course is described in Chapter 61.

Lesions of the intercostal nerves

Subluxation of the interchondral joints between the lower costal cartilages may trap the intercostal nerves, causing referred abdominal pain. The dorsal cutaneous branch of an intercostal nerve can become entrapped as it penetrates the fascia of erector spinae. This produces an area of numbness, usually with painful paraesthesia, which extends approximately 10 cm from the midline laterally and 10 cm in length (notalgia paraesthetica); the area between the medial edge of the scapula and the spine is commonly affected. The anterior cutaneous branches of the intercostal nerves can also become entrapped as they penetrate the fascia of rectus abdominis, and this produces an area of numbness on the abdomen, usually with painful paraesthesia, which extends from the midline laterally for 10 or 12 cm (rectus abdominis syndrome).

Twelfth thoracic ventral ramus (subcostal nerve)

The ventral ramus of the twelfth thoracic nerve (subcostal nerve) is larger than the others and gives a communicating branch to the first lumbar ventral ramus (sometimes termed the dorsolumbar nerve). Like the intercostal nerves, it soon gives off a collateral branch and then accompanies the subcostal vessels along the inferior border of the twelfth rib, passing behind the lateral arcuate ligament and kidney, and in front of the upper part of quadratus lumborum. It perforates the aponeurosis of the origin of transversus abdominis and passes forwards between that muscle and internal oblique, to be distributed in the same manner as the lower intercostal nerves. The subcostal nerve connects with the iliohypogastric nerve of the lumbar plexus and sends a branch to pyramidalis. Its lateral cutaneous branch pierces the internal and external oblique muscles and supplies the lowest slip of the latter. It usually descends over the iliac crest 5 cm behind the anterior superior iliac spine (see Fig. 61.5) and is distributed to the anterior gluteal skin; some filaments reach as low as the greater trochanter of the femur.

THORACIC DORSAL SPINAL RAMI

Thoracic dorsal rami pass backwards close to the vertebral zygapophysial (facet) joints and divide into medial and lateral branches. The medial branch emerges between the joint and the medial edge of the superior costotransverse ligament and intertransverse muscle. The lateral branch runs in the interval between the ligament and the muscle before inclining posteriorly on the medial side of levator costae.

Medial branches of the upper six thoracic dorsal rami pass between and supply semispinalis thoracis and multifidus; they then pierce the rhomboids and trapezius and reach the skin near the vertebral spines (see Fig. 45.8). Medial branches of the lower six thoracic dorsal rami are distributed mainly to multifidus and longissimus thoracis; occasionally, they give filaments to the skin in the median region. Lateral branches increase in size from above downwards. They run through or deep to longissimus thoracis to the interval between it and iliocostalis cervicis, supplying these muscles and levatores costarum; the lower five or six also give off cutaneous branches, which pierce serratus posterior inferior and latissimus dorsi in line with the costal angles. The lateral branches of a variable number of the upper thoracic rami also supply the skin. The lateral branch of the twelfth dorsal ramus sends a filament medially along the iliac crest and then passes inferiorly to innervate the skin of the anterior part of the gluteal region.

Medial cutaneous branches of the thoracic dorsal rami descend for some distance close to the vertebral spines before reaching the skin. Lateral branches descend for a considerable distance, which may be as much as the breadth of four ribs, before they become superficial, e.g. the branch of the twelfth thoracic ramus reaches the skin only a little way above the iliac crest.

BREAST

The breasts form a secondary sexual feature of females and are a source of nutrition for the neonate. In young adult females, each breast is a rounded eminence largely lying within the superficial fascia anterior to the upper thorax but spreading laterally to a variable extent (**Figs 53.22, 53.23A**). Breast shape and size depend on genetic, racial and dietary factors and on the age, parity and menopausal status of the individual. Breasts may be hemispherical, conical, variably pendulous, piriform or thin and flattened. In the adult female, the base of the breast, i.e. its attached surface, extends vertically from the second or third to the sixth rib, and in the transverse plane from the sternal edge medially almost to the mid-axillary line laterally. The superolateral quadrant is prolonged towards the axilla along the inferolateral edge of pectoralis major, from which it projects a little, and may extend through the deep fascia up to the apex of the axilla (the axillary tail of Spence). The trunk superficial fascial system splits to enclose the breast to form the anterior and posterior lamellae. Posterior extensions of the superficial fascial system connect the breast to the pectoralis fascia, part of the deep fascial system. The inframammary crease is a zone of adherence of the superficial fascial system to the underlying chest wall at the inferior crescent of the breast.

The breast lies on the deep pectoral fascia, which in turn overlies pectoralis major and serratus anterior superiorly and external oblique

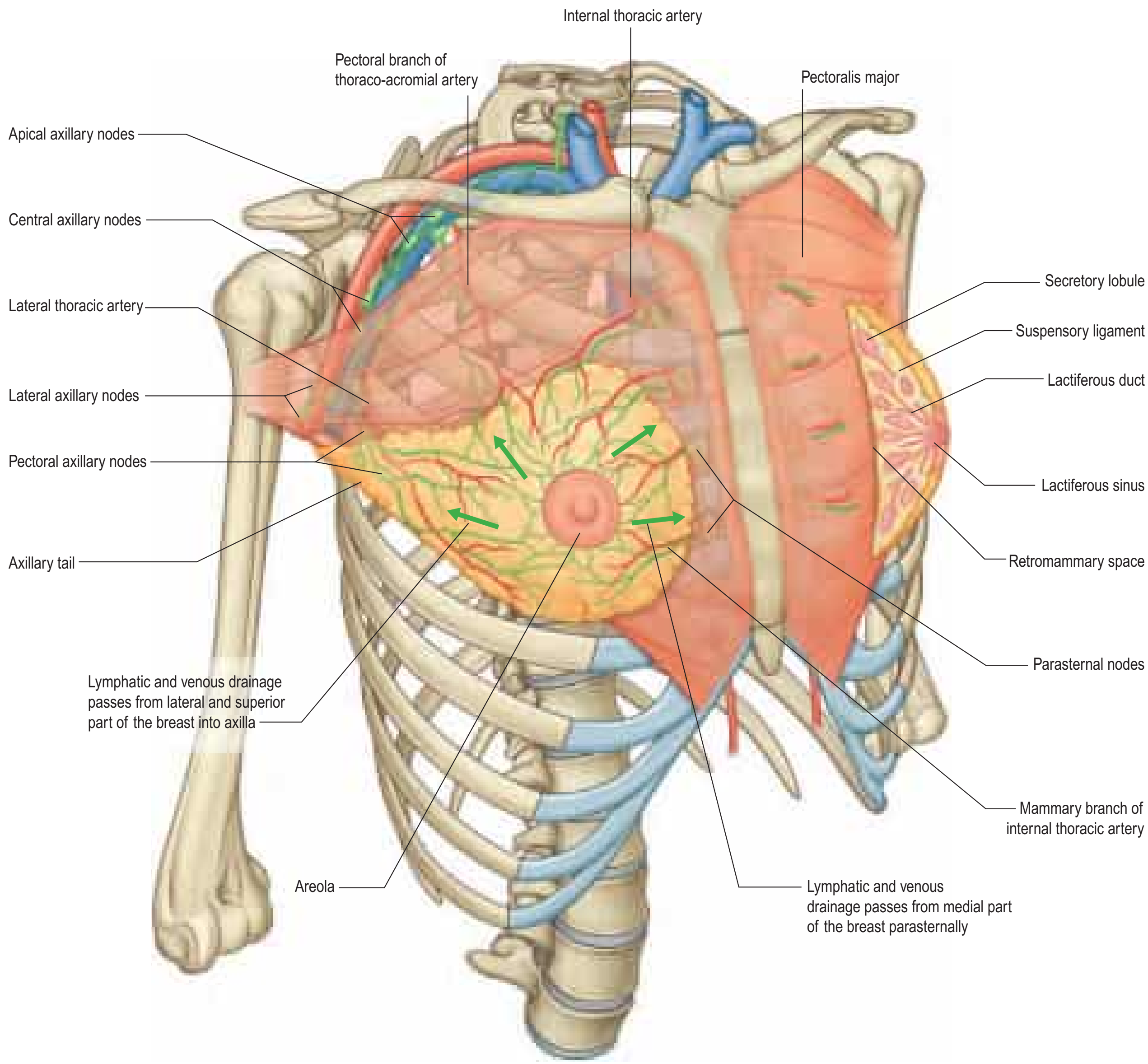

Fig. 53.22 The relations of the breast. (With permission from Drake, RL, Vogl, AW, Mitchell, A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

SECTION 7

and its aponeurosis inferiorly, as the latter forms the anterior wall of the rectus sheath. Between the breast and the deep fascia, the loose connective tissue in the 'submammary space' allows the breast some degree of movement on the deep pectoral fascia. Advanced mammary carcinoma may cause tethering or fixation of the breast to the underlying musculature. Occasionally, small projections of glandular tissue may pass through the deep fascia into the underlying muscle in normal subjects.

NIPPLE AND AREOLA

The nipple projects from the centre of the breast anteriorly (Fig. 53.23A–C). It may be cylindrical and rounded, hemispherical or flattened, depending on the effects of developmental, nervous or hormonal factors and external temperature on the erectile properties of the subareolar muscle of the nipple. The level of the nipple varies widely. In females, its site is dependent on the size and shape of the breasts; it overlies the fourth intercostal space in most young women. In the male, the nipple is usually sited in the fourth intercostal space in the mid-clavicular line. In the young adult of either sex, the nipples are usually positioned 20–23 cm from the suprasternal notch in the mid-clavicular line and 20–23 cm apart in the horizontal plane. With increasing age and parity, female breasts adopt a more ptotic shape and the position of the nipple drops to the level of the inframammary crease or below. In the nulliparous, the nipple is pink, light brown or darker, depending on the general melanization of the body. Occasionally, the nipple may not evert during prenatal development and it remains permanently retracted (see below).

The skin covering the nipple and the surrounding areola (the disc of skin that circles the base of the nipple) has a convoluted surface. It contains numerous sweat and sebaceous glands that open directly on to the skin surface. The oily secretion of these specialized sebaceous glands acts as a protective lubricant and facilitates latching of the neonate during lactation; the glands are often visible in parous women, arranged circumferentially as small elevations, Montgomery's tubercles, around the areola close to the margin. Other areolar glands, intermediate in structure between mammary and sweat glands, become enlarged in pregnancy and lactation as subcutaneous tubercles. The sebaceous glands of the areola are not usually associated with hair follicles. The skin of the nipple and areola is rich in melanocytes and is therefore typically darker than the skin covering the remainder of the breast; further darkening occurs during the second month of pregnancy, and subsequently persists to a variable degree.

SOFT TISSUE

The breasts are composed of lobes that contain a network of glandular tissue consisting of branching ducts and terminal secretory lobules in a connective tissue stroma (see Fig. 53.30). The terminal duct lobular unit is the functional milk secretory component of the breast; pathologically, it gives rise to primary malignant lesions within the breast. Although the lobes are usually described as discrete territories, they intertwine in three dimensions and merge at their edges; they cannot be distinguished during surgery. The connective tissue stroma that surrounds the lobules is dense and fibrocollagenous, whereas intralobular connective tissue has a loose texture that allows the rapid expansion of

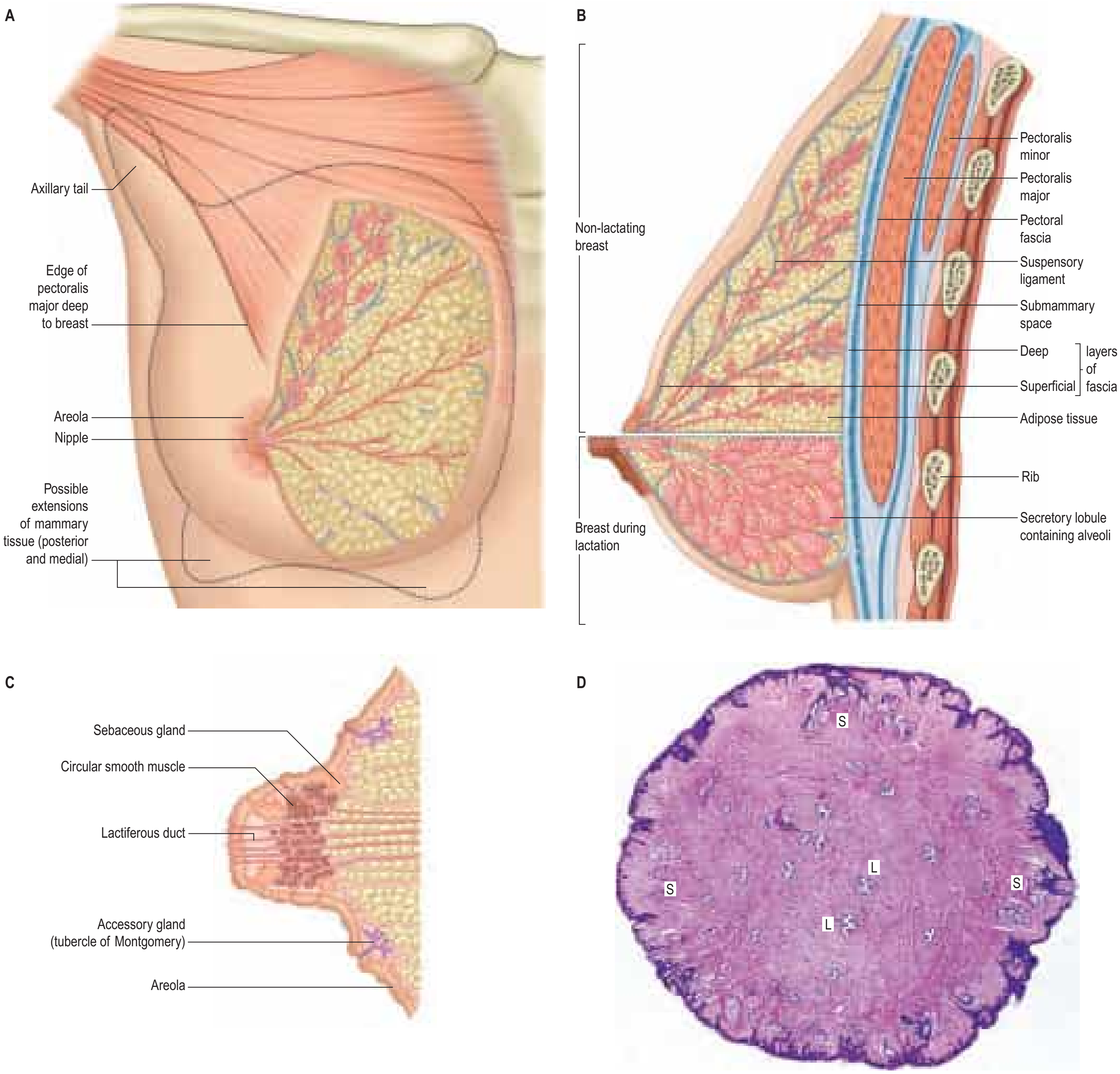

Fig. 53.23 **A,** The structure of the breast. **B,** Changes in the breast during lactation. **C,** A section of the nipple. **D,** A cross-section of the nipple. There is a corrugated layer of stratified squamous keratinized epithelium over the nipple surface; 20 or more lactiferous ducts (L) open on to the surface; sebaceous glands (S) are deep to the epidermis. (**D,** With permission from Dr JB Kerr, Monash University, from Kerr JB 1999 Atlas of Functional Histology. London: Mosby.)

secretory tissue during pregnancy (see Fig. 53.30B). Fibrous strands or sheets consisting of condensations of connective tissue extend between the layer of deep fascia that covers the muscles of the anterior chest wall and the dermis. These suspensory ligaments (of Astley Cooper) are often well developed in the upper part of the breast and support the breast tissue, helping to maintain its non-ptotic form. Elsewhere in the normal breast, fibrous tissue surrounds the glandular components and extends to the skin and nipple, assisting the mechanical coherence of the gland. The interlobar stroma contains variable amounts of adipose tissue, which is responsible for much of the increase in breast size at puberty.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Arteries

The breasts are supplied by branches of the axillary, internal thoracic and some intercostal arteries. The axillary artery supplies blood via the superior thoracic artery, the pectoral branches of the thoraco-acromial artery, the lateral thoracic artery (via branches that curve around the lateral border of pectoralis major to supply the lateral aspect of the breast) and the subscapular artery. The internal thoracic artery supplies perforating branches to the anteromedial part of the breast. The second to fourth anterior intercostal arteries supply perforating branches more laterally in the anterior thorax. The second perforating artery is usually the largest and supplies the upper region of the breast, the nipple, areola and adjacent breast tissue.

Veins

Blood drains from the circular venous plexus around the areola and from the glandular tissue of the breast into the axillary, internal thoracic and intercostal veins via veins that accompany the corresponding arteries. Individual variation is common.

Lymphatic drainage

The lymphatic flow of the breast is of great clinical significance because metastatic dissemination occurs principally by the lymphatic routes (the breasts are the site of malignant change in as many as 1 in 8 women). The dominant lymphatic drainage of the breast is derived from the dermal network. The breast lymphatics branch extensively and do not contain valves; lymphatic blockage through tumour occlusion may therefore result in reverse blood flow through the lymphatic channels. The direction of lymphatic flow within the breast parallels the major venous tributaries and enters the regional lymph nodes via the extensive periductal and perilobular network of lymphatic channels. Most of these lymphatics drain into the axillary group of regional lymph nodes either directly or through the retro-areolar lymphatic plexus (see Fig. 48.48). Dermal lymphatics also penetrate pectoralis major to join channels that drain the deeper parenchymal tissues, and then follow the vascular channels to terminate in the subclavicular lymph nodes.

Lymphatics from the left breast ultimately terminate in the thoracic duct and, subsequently, the left subclavian vein. On the right, the lymphatics ultimately drain into the right subclavian vein near its junction with the internal jugular vein. Part of the medial side of the right breast drains towards the internal thoracic group of lymph nodes. The internal thoracic chain may drain inferiorly via the superior and inferior epigastric lymphatic routes to the groin. Connecting lymphatics across the midline may provide access of lymphatic flow to the opposite axilla.

Axillary nodes receive more than 75% of the lymph from the breast (**Fig. 53.24**). There are 20–40 nodes, grouped artificially as pectoral (anterior), subscapular (posterior), central and apical. Surgically, the nodes are described in relation to pectoralis minor. Those lying below pectoralis minor are the low nodes (level 1), those behind the muscle are the middle group (level 2), while the nodes between the upper border of pectoralis minor and the lower border of the clavicle are the upper or apical nodes (level 3). There may be one or two other nodes between pectoralis minor and major; this interpectoral group of nodes are also known as Rotter's nodes. Efferent vessels directly from the breast pass round the anterior axillary border through the axillary fascia to the pectoral lymph nodes; some may pass directly to the subscapular nodes. A few vessels pass from the superior part of the breast to the

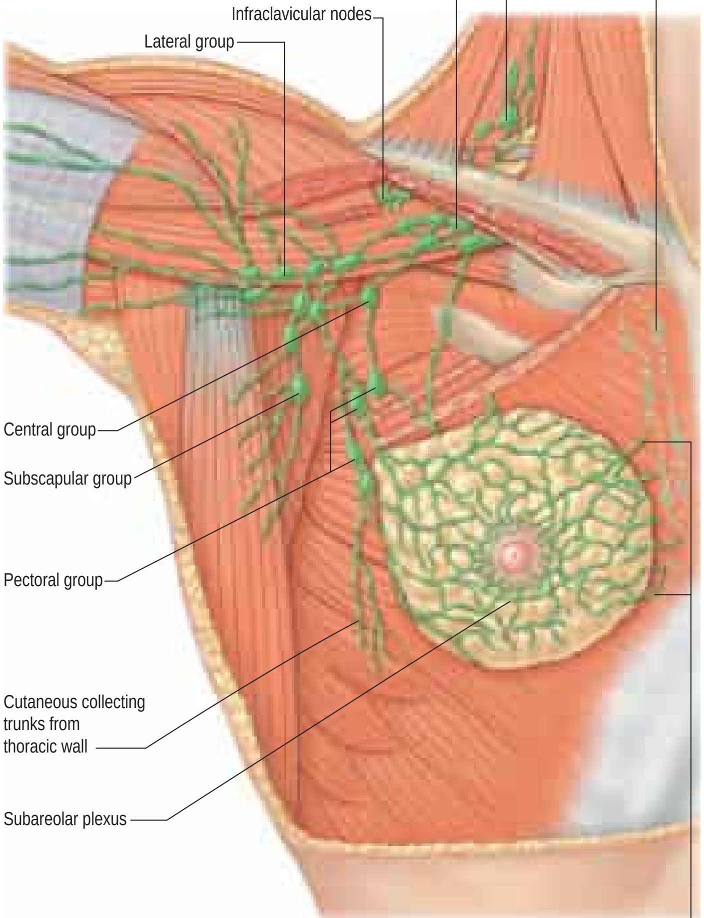

Fig. 53.24 Lymph vessels of the breast and the draining axillary and supraclavicular nodal groups.

apical axillary nodes, sometimes interrupted by the infraclavicular nodes or by small, inconstant, interpectoral nodes. Most of the remainder drain to parasternal nodes from the medial and lateral parts of the breast; they accompany perforating branches of the internal thoracic artery. Lymphatic vessels occasionally follow lateral cutaneous branches of the posterior intercostal arteries to the intercostal nodes.

Axillary surgery in breast cancer

Axillary lymph node dissection may be performed because the presence of metastases within axillary lymph nodes has strong prognostic significance and might influence decisions on adjuvant therapy. However, axillary lymph node dissection can lead to chronic postoperative problems such as pain, seroma formation, reduced mobility of the arm, impaired sensation and lymphoedema. The vessels and nerves have to be carefully identified at surgery as anatomical landmarks.

Lymphatic drainage in breast cancer and role of sentinel lymph node biopsy

Lymphatic mapping with sentinel lymph node biopsy has become an important technique in the staging of patients with early breast cancer. A radiolabelled colloid is injected into either the subareolar tissue of the index quadrant of the breast or the peritumoral and intradermal tissue overlying the primary breast cancer. At the time of surgery, a vital blue dye is injected after general anaesthesia is established. The combination of radioisotope and dye provides the most accurate means of localizing the sentinel node (Tanis et al 2001). The latter represents the first draining node of the axilla and is surgically removed for careful histopathological analysis to detect the presence of metastases.

INNERVATION

The breast is innervated by anterior and lateral branches of the fourth to sixth intercostal nerves, which carry sensory and sympathetic efferent fibres. The nipple is supplied from the anterior branch of the lateral cutaneous branch of T4, which forms an extensive plexus within the nipple; its sensory fibres terminate close to the epithelium as free endings, Meissner corpuscles and Merkel disc endings. These are essential in signalling suckling to the central nervous system. Secretory activities of the gland are largely controlled by ovarian and hypophysial hormones rather than by efferent motor fibres. The areola has fewer sensory endings.

MICROSTRUCTURE

The microstructure of breast tissue varies with age, time in the menstrual cycle, pregnancy and lactation. The following description relates to the mature, resting breast. For most of their lengths, the ducts are lined by columnar epithelium (**Fig. 53.25A**). In the larger ducts, this is two cells thick but, in the smaller ones, only a single layer of columnar or cuboidal cells is present. The bases of these cells are in close contact with numerous myoepithelial cells of ectodermal origin, similar to those of certain other glandular epithelia (see Figs 2.3, 2.4, 53.30A). Myoepithelial cells are so numerous that they form a distinct layer surrounding the ducts and presumptive alveoli, and give the epithelium a bilayered appearance.

Lactiferous ducts draining each lobe of the breast pass through the nipple and open on to its tip as 15–20 orifices. Near its orifice, each of these ducts is slightly expanded as a lactiferous sinus, which, in the lactating breast, is further dilated by the presence of milk. Each lactiferous duct is therefore connected to a system of ducts and lobules, surrounded by connective tissue stroma, collectively forming a lobe of the breast. Lobules consist of the portions of the glands that have secretory potential. Their structure varies according to hormonal status. In the mature resting breast, each lobule consists of a cluster of blind-ended, branched ductules (**Fig. 53.25B**), whose termini lack mature terminal alveoli (acini), which are the sites of milk secretion in the lactating breast (see Fig. 53.30B). The stratified cuboidal lining is replaced by keratinized stratified squamous epithelium, continuous with the epidermis, close to the openings of the lactiferous ducts on the nipple. Shed squames may sometimes block the duct apertures in the non-pregnant breast.

Internally, the nipple is composed mostly of collagenous dense connective tissue and contains numerous elastic fibres that wrinkle the overlying skin. Deep to the nipple and areola, bundles of smooth muscle cells are arranged radially and circumferentially within the connective tissue and are thought to be remnants of the panniculus carnosus. Their contraction, induced by cold or tactile stimuli (e.g. in

If the histopathological results prove negative, the morbidity associated with axillary dissection can be avoided. In women who are found to have metastases to the axillary nodal group, axillary completion dissection or radiotherapy is offered. The majority of sentinel nodes are found in the low axilla (level 1). Higher-level 'skip' nodes are involved in 5–10% of cases. Occasionally, the sentinel node is identified in an extra-axillary position, either within the breast parenchyma as an intramammary lymph node, or as an internal thoracic lymph node or in the supraclavicular fossa.

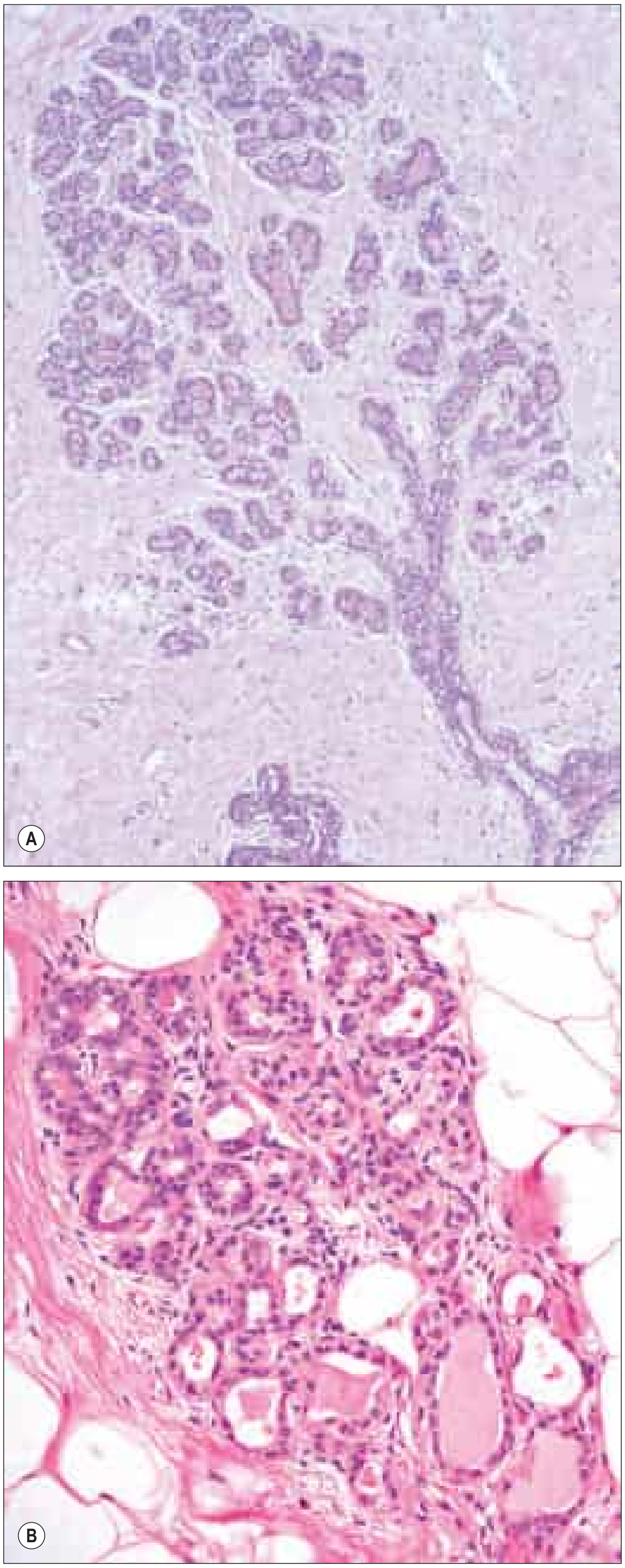

Fig. 53.25 **A,** A glandular lobule surrounded by collagenous interlobular connective tissue in the mature resting breast. A terminal duct (bottom right) branches extensively to terminate in rudimentary acini. **B,** The rudimentary acini shown at higher magnification, surrounded by fibrous and adipose connective tissue. (**A,** With permission from Young B, Heath JW 2000 Wheater's Functional Histology. Edinburgh: Churchill Livingstone. **B,** Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

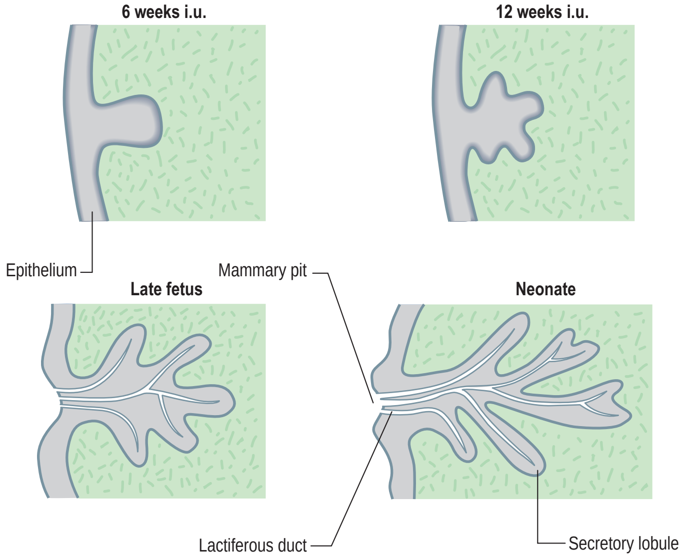

Fig. 53.27 Early development of breast epithelium. Abbreviation: i.u., intrauterine.

suckling), causes erection of the nipple and wrinkling of the surrounding areola.

BREAST CANCER

Available with the *Gray's Anatomy* e-book

DEVELOPMENT

The epithelial/mesenchymal interactions that will give rise to the glandular tissue of the breast in both sexes can first be seen at about the fifth or sixth week, when two ventral bands of thickened ectoderm, the mammary ridges or milk lines, extend from the axilla to the inguinal region. Usually, invagination of the thoracic mammary bud occurs by day 49, and the remaining mammary line involutes.

The thoracic ectodermal ingrowths branch into 15–20 solid buds of ectoderm that will become the lactiferous ducts and their associated lobes of alveoli in the fully formed gland. They are surrounded by somatopleuric mesenchyme that forms the connective tissue, fat and vasculature that is invaded by the mammary nerves. Continued cell proliferation, elongation and further branching produce the alveoli and define the duct system. Nipple formation begins at day 56, primitive ducts (mammary sprouts) develop at 84 days, and canalization occurs at about the 150th day. During the last 2 months of gestation, the ducts become canalized and the epidermis at the point of original development of the gland forms a small mammary pit, into which the lactiferous tubules open (Fig. 53.27). Perinatally, the nipple is formed by mesenchymal proliferation. Should this process fail, the ducts open into shallow pits. Rarely, the nipple may not develop (athelia), a phenomenon that occurs more commonly in accessory breast tissue.

At birth, the breasts have reached a similar developmental stage in both sexes, and the combination of fetal prolactin and maternal oestrogen may give rise to transient hyperplasia and secretion of 'witch's milk'. In males, the breasts normally remain undeveloped whereas in females at puberty, in late pregnancy and during the period of lactation, they undergo further, hormone-dependent developmental changes. The female breast is a unique organ in that it remains in a rudimentary (i.e. fetal) form until puberty, at which time its development continues under the influence of sex hormones. The adult form is reached in late adolescence, i.e. the breast is then able to function as a milk-producing organ. At menopause, the breast involutes into a predominantly fatty organ with minimal glandular parenchyma.

Accessory breast tissue and nipples

Polymastia (supernumerary breasts) and polythelia (supernumerary nipples) may develop in males and females anywhere along the length of the mammary ridges (milk lines; Fig. 53.28). Conversely, breast tissue may not develop at all (amastia), or there may be nipple development but no breast tissue (amazia). Supernumerary breast development

Breast cancer is a common disease, particularly in postmenopausal women (see Fentiman 1993). Each year, in the United Kingdom, there are approximately 40,000 new cases diagnosed and 14,000 deaths. Male breast cancers constitute up to 1% of all mammary malignancies and may include tissue beyond the areolar boundary.

Breast cancers arise within the epithelia of lobules or ducts. As they increase in size and infiltrate the stroma, they often lead to a fibrous tissue reaction. Breast lumps may be classified as benign or malignant masses. Malignant breast lumps may have the clinical signs of infiltrating adjacent structures, leading to a hard and irregular mass with skin-tethering, muscle fixation, skin infiltration or oedema of the overlying skin (*peau d'orange*). Standard radiological imaging investigations of the breast include mammography and ultrasonography. Mammography is useful in detecting the presence of malignant masses as a stellate opacity with architectural distortion of the surrounding parenchyma. Early stages of preinvasive breast cancer, ductal cancer *in situ*, may appear as microcalcification on the mammogram. An ultrasound scan of the breast is helpful to distinguish solid from cystic (or fluid-filled) masses, and can also aid in the distinction of breast cancer from benign lumps by the different attenuation characteristics of the ultrasound waves and blood-flow patterns. Wrinkles caused by contraction of muscular fibres in the areola described above may create critical angle shadowing in breast ultrasound.

Magnetic resonance imaging (MRI) is also useful in diagnosis, particularly in women with younger, denser breasts, achieving higher sensitivity of breast cancer detection but with lower specificity. A cytological diagnosis of a breast lump can be achieved using fine needle aspiration (FNA) to evaluate the cellular component of a lump. FNA is also useful to drain symptomatic breast cysts. A wide-bore needle biopsy is a procedure performed under local anaesthetic to obtain several pieces of specimen for a histological diagnosis with the tissue architecture maintained. This permits distinction between *in situ* and invasive cancer, and provides an indication of tumour subtype by evaluation of the pathological characteristics. Vacuum-assisted mammotome biopsies based on the wide-bore needle enable some benign and indeterminate lesions to be excised for full histological analysis and, in some patients, can avoid the need for conventional open surgery.

If a breast lump has to be surgically removed, the incision should be based, whenever possible, in the relaxed skin tension lines, for the best cosmetic results. In women with sizeable malignant lesions in whom breast conservation surgery is to be attempted, the skin incision should be planned with consideration of the possible requirement for a subsequent mastectomy if the margins of excision are incompletely excised by pathological criteria (Swanson et al 2002). Most women with single breast cancers of up to 4 cm in diameter are treated by breast conservation rather than mastectomy. This is a combination of surgery (tumour excision and sentinel node biopsy or clearance) together with external beam radiotherapy. Patients with larger tumours are treated by modified radical mastectomy with axillary lymph node clearance. During axillary dissection to clear the axillary lymph nodes, the nerve to serratus anterior, the thoracodorsal vessels, the nerve to latissimus dorsi, and the medial and lateral pectoral nerves are all identified and carefully preserved. The boundaries of the axillary dissection are the nerve to serratus anterior medially, the thoracodorsal pedicle laterally and the axillary vein superiorly. The posterior limit of the dissection is the ventral surface of subscapularis. The superomedial limit of a level 1 axillary dissection extends to the lateral border of pectoralis minor at the apex of the axilla; a level 2 dissection extends to the medial border of pectoralis minor; and a level 3 dissection extends beyond the medial border until it reaches the point where the axilla is limited by the first rib (the latter is easily distinguished by its flat lateral surface, easily palpable at surgery). Failure to preserve the nerve to serratus anterior will result in winging of the scapula. The intercostobrachial nerve is often sacrificed in an axillary lymph node clearance operation, and this may result in anaesthesia of a narrow strip of sensation in the upper medial border of the arm.

Reconstructive surgery for breast cancer disease

Breast reconstruction may be performed at the time of mastectomy for breast cancer, or at a later stage (Serletti and Moran 2000). During a mastectomy where an immediate breast reconstruction is planned, the glandular breast tissue is removed either through a conventional mastectomy skin incision or in association with preservation of the native breast skin, a form of mastectomy termed a skin-sparing mastectomy. Patient selection is important to minimize risks of breast cancer recurrence in the skin of the reconstructed breast mound or associated lymph nodes. In an implant-based immediate breast reconstruction, a tissue expander is placed beneath the musculofascial plane consisting of pectoralis major, in continuity laterally under serratus anterior with the intervening fascia, and the abdominal fascia inferiorly. The lower fibres of the attachment of pectoralis major to the sixth rib need to be detached; otherwise, the implant will be placed too high on the chest wall. The tissue expander has to stretch the potential space below this musculofascial plane to match the space created by the removed breast beneath the subcutaneous plane and the deep fascia overlying pectoralis major. The tissue expander is subsequently replaced by a silicone or saline breast implant.

The skin and volume component of the breast can also be replaced by the latissimus dorsi musculocutaneous flap, with or without an underlying silicone or saline breast implant. Latissimus dorsi and an overlying paddle of skin are raised on its dominant vascular pedicle: namely, the thoracodorsal vessels, which enter the muscle on its deep surface and send perforating branches through the muscle to the overlying skin. The flap is transferred in an arc from the back to the front of the chest and fashioned to make a new breast. An implant may be required to provide additional volume replacement, in which case it is placed beneath the flap. The donor defect on the back is directly sutured. In selected cases, this flap can be raised with additional superficial fascia and fat, to avoid the need for a silicone implant. This is known as the extended or autologous latissimus dorsi flap.

Women with a suitable abdominal panniculus may have breast reconstruction utilizing this excess tissue. Historically, this was achieved with a pedicled transverse rectus abdominis musculocutaneous (TRAM) flap based on the superior epigastric artery, where the skin and fat of the lower abdomen, attached to the rectus muscle, are transposed into the breast defect and shaped into a breast reconstruction.

The development of microvascular techniques over the last 30 years has seen an increase in the popularity of free tissue transfers, based on the deep inferior epigastric vessels, including muscle-sparing (MS-) TRAM flaps and deep inferior epigastric artery perforator (DIEP) flaps. These flaps require dissection of the blood vessels within rectus abdominis, rather than sacrifice of the entire muscle, and this aims to reduce donor site morbidity, including hernias and bulging. Radiological investigations (CT angiograms, MRI and duplex) are used to determine the calibre and position of perforators within these flaps to improve the accuracy of dissection and enhance surgical outcomes (Fig. 53.26). Free TRAM and DIEP flap vessels are anastomosed to recipient vessels in the chest; the internal thoracic or the thoracodorsal artery is used most frequently for this purpose.

Breast reconstruction using an autologous method, such as the extended latissimus dorsi, free MS-TRAM and DIEP flap, avoids the need for a silicone implant and has advantages in terms of producing a more natural appearance and achieving greater longevity of results. Free tissue transfers for breast reconstruction are also performed using buttock tissue (superior and inferior gluteal artery perforator flaps) and thigh tissue (transverse upper gracilis and anterolateral thigh flaps).

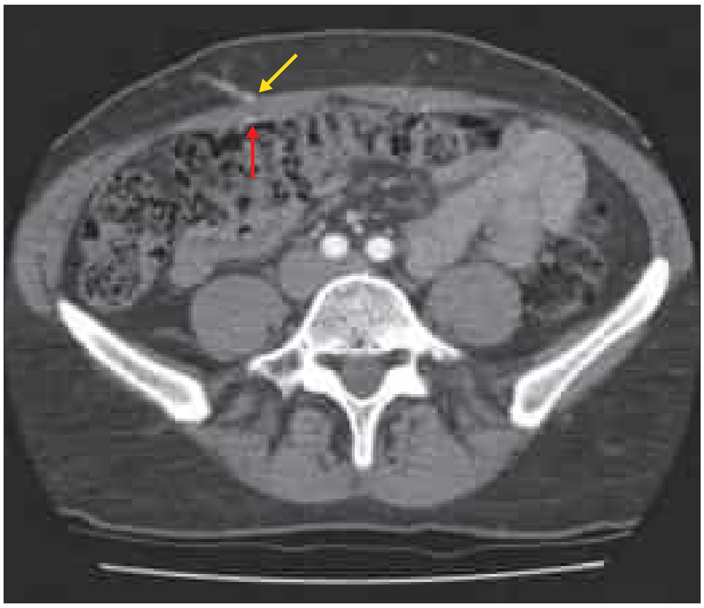

Fig. 53.26 A CT angiogram showing the deep inferior epigastric artery (red arrow) and a perforator (yellow arrow) emerging from the rectus sheath for MS-TRAM and DIEP flap reconstruction.

CHAPTER 53

Breast reconstruction using autologous tissue

One of the earliest recorded breast reconstructions using latissimus dorsi, based on the thoracodorsal pedicle, was published in 1906 (Tansini 1906). The flap incorporated superficial fascia and a skin paddle attached to the muscle, and was used to replace part of the breast skin envelope that had been removed during the mastectomy. This form of breast reconstruction inexplicably fell out of regular use until the 1970s, when there was renewed interest in the latissimus dorsi musculocutaneous flap combined with silicone breast implants (Schneider et al 1978, Bostwick 1977) and improved definitions of the vascular supply and territory of the flap (McCraw et al 1977, Maxwell et al 1979).

Later in the twentieth century, further developments of this most versatile flap meant that breast reconstruction could be performed without a silicone implant. The so-called 'extended' flap (Hokin and Silfverskiold 1987) used autologous tissues exclusively; it was designed to satisfy an increasing demand for breast reconstruction in an era when silicone implants were under scrutiny because of safety fears and proposed links with autoimmune diseases. Further developments to maximize the volume available for autologous reconstructions of small to medium-sized breasts included harvesting larger skin paddles from the back and burying them beneath the breast skin envelope after de-epithelializing their surface (Papp et al 1988), and inclusion of additional fat pads in the back to increase flap volume (Delay et al 1998).

Despite the enhancements to the latissimus dorsi flap, the lower abdominal region provides a plentiful and more reliable source of tissue for reconstructive surgeons and also avoids the sometimes troublesome and often unsightly dorsal donor site wound. DIEP or MS-TRAM flaps are generally considered the first choice for breast reconstruction when the tissues are adequate.

The first description of lower abdominal tissues being used as a free flap appeared in 1979 (Holmstrom 1979), although at this time the pedicled TRAM flap was a far more popular choice for reconstruction. This was mainly because microvascular techniques were still in development and the required instruments and microscopes were not widely available. However, over the years, it became possible to achieve excellent, reliable results without the problems of fat necrosis and abdominal wall weakness associated with the pedicled version (Schusterman 1994, Grotting 1989, Arnez et al 1991). The TRAM flap particularly lends itself to microsurgical transfer on account of the characteristics of the supplying vessels (deep inferior epigastric artery is 2.5–3 mm in diameter and its venae comitantes are 3 mm in diameter); the length of the pedicle (frequently up to 15 cm); and the large surface area of skin supplied by these vessels (from the midline to a point lateral to the anterior superior iliac spine below the level of the umbilicus).

Further developments in dissection techniques, combined with a desire to reduce donor site morbidity, led to the first description of a 'perforator' flap based on the same vessels (Allen and Treece 1994). This is the DIEP flap, which captures the same soft tissue area of the abdomen for reconstruction, but attempts to avoid injury to rectus abdominis by dissecting the perforating vessels through the muscle rather than by sacrificing it. Some authors have identified fewer donor site bulges and herniae with DIEP flaps compared with TRAM flaps (Blondeel et al 1997), and it is for this reason that DIEP flaps are the preferred choice, if possible. However, the perforating vessels are not always anatomically favourable for allowing safe harvest of the flap. Many clinicians have adopted radiological studies (CT angiogram, duplex or MR angiogram) to determine the calibre and position of the vessels preoperatively for more accurate surgical planning (see Fig. 53.26).

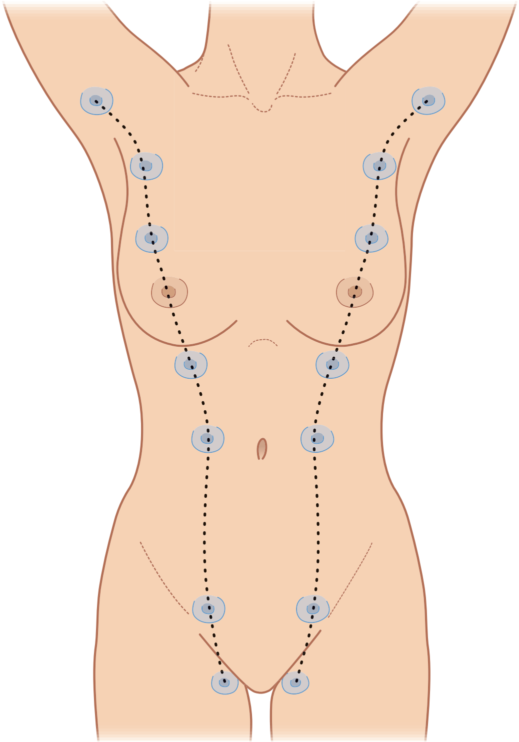

Fig. 53.28 Milk lines.

occurs in most cases in the thoracic region, just inferior to the normal breast (90%), but may also occur in the axillary (5%) and abdominal regions (5%) and even on to the medial thigh. Polythelia occurs along the same mammary line but no underlying glandular tissue develops. This condition is rare in the female population but it is more common in males, in whom the accessory nipple may be mistaken for a mole.

Congenital inversion of nipple

Congenital inversion of the nipples occurs rarely in the female population and is almost always bilateral. The majority of the cases are umbilicated, i.e. the nipple can be easily pulled forwards from its depressed position underneath the areolar surface. The condition is thought to be due to failure of proliferation of the mesenchymal tissue, which fails to push the nipple out. The remaining cases are due to invagination of the nipple. Apart from psychological implications, inversion of the nipple may cause recurrent mastitis and difficulty with breast feeding. The abnormality may be corrected surgically.

AGE-RELATED CHANGES

Prepuberty

The neonatal breast contains lactiferous ducts but no alveoli. Until puberty, little branching of the ducts occurs, and any slight mammary enlargement reflects the growth of fibrous stroma and fat.

Puberty

In the postpubertal female, the ducts become branched on stimulation by ovarian oestrogens. The ends of the branches form solid, spheroidal masses of granular polyhedral cells: the potential alveoli. Oestrogens also promote adipocyte differentiation from mesenchymal cells in the interlobar stroma. Breast enlargement at puberty is mainly a consequence of lipid accumulation by these adipocytes.

From puberty onwards, externally recognizable breast development (thelarche) can be divided into five separate phases (**Fig. 53.29**): elevation of the breast bud (phase I); glandular subareolar tissue is present and both nipple and breast project from the chest wall as a single mass (phase II); the areola increases in diameter and becomes pigmented, and there is proliferation of palpable breast tissue (phase III); further pigmentation and enlargement occur in the areola, so that the nipple and areola form a secondary mass anterior to the main part of the breast (phase IV); and a smooth contour to the breast develops (phase V).

Changes during the menstrual cycle

Changes occur in the breast tissues in the menstrual cycle. In the follicular phase (days 3–14), the stroma becomes less dense. Various changes, including luminal expansion, take place in the ducts; there are occasional mitoses but no secretion. In the luteal phase (days 15–28), there is a progressive increase in stromal density and the ducts have an open lumen that contains secretion, associated with flattening of the epithelial cells. Cell proliferation is maximal on day 26 and thereafter the ductal system undergoes reduction; epithelial cell apoptosis is greatest on day 28 of the cycle. There are also changes in blood flow, which are greatest at mid-cycle, and an increase in the water content of the stroma in the second half of the menstrual cycle.

Postmenopausal

Progressive atrophy of lobules and ducts occurs after the menopause, and there is fatty replacement of glandular breast tissue. A few ducts may remain. The stroma becomes much less cellular and collagenous fibres decrease. The amount of adipose tissue varies widely between individuals, and the breast may return to a condition similar to the prepubertal state.

CHANGES ASSOCIATED WITH PREGNANCY AND LACTATION

Pregnancy

As the output of oestrogen and progesterone, produced first by the corpus luteum and later by the placenta, rises during pregnancy, the intralobular ductal epithelium proliferates and the cells increase in size; the number and length of the ductal branches therefore increase. Alveoli develop at their termini and expand as their cells and lumina fill with newly synthesized and secreted milk. The myoepithelial cells, which are initially spindle-shaped, become highly branched stellate cells, especially around the alveoli. Adjacent myoepithelial cells intermesh to form a basket-like network around the alveoli and ducts, interposed between the basal lamina and the luminal cells (see Fig. 2.3). Their cytoplasm contains actin and myosin filaments, and they are contractile (see Fig. 2.4). There is a concomitant reduction in adipose tissue in the stroma. The numbers of lymphocytes, including plasma cells, and eosinophils increase greatly. Blood flow through the breast increases.

Secretory activity in the alveolar cells rises progressively in the latter half of pregnancy. In late pregnancy, and for a few days after parturition, their product is different from later milk and is known as colostrum, characteristically low in lipid but rich in protein and immunoglobulins. Proliferation of the glandular breast parenchyma results in an overall increase in breast size through gestation.

Lactation

True milk secretion begins a few days after parturition as a result of a reduction in circulating oestrogen and progesterone, a change that appears to stimulate production of prolactin by the anterior hypophysis. Milk distends the alveoli so that the cells flatten as secretion increases (see Fig. 53.23B; **Fig. 53.30**). The alveolar cell cytoplasm accumulates membrane-bound granules of casein and other milk proteins, and these are released from the apical plasma membrane by membrane fusion (merocrine secretion; see Fig. 2.6). Lipid vacuoles are formed directly in the apical cytoplasm as small lipid droplets that fuse with each other to create large 'milk vacuoles' up to 10 μm across. These are released as intact lipid droplets with a thin surround of apical plasma membrane and adjacent cytoplasm (apocrine secretion; see Fig. 2.6). On hormonal stimulation by oxytocin, myoepithelial cells contract to expel alveolar secretions into the ductal system in readiness for suckling.

After the onset of lactation, there is a gradual reduction in the numbers of lymphocytes and eosinophils in the stroma, although plasma cells continue to synthesize immunoglobulin A (IgA) for secretion into the milk. Alveolar cells take up IgA synthesized by adjacent plasma cells by endocytosis at their basal surfaces and secrete it apically, as dimers complexed to the epithelial secretory component.

Post lactation

When lactation ceases, which may be after as long as 3½ years, the secretory tissue undergoes some involution but the ducts and alveoli never return completely to the pre-pregnant state. Two major processes

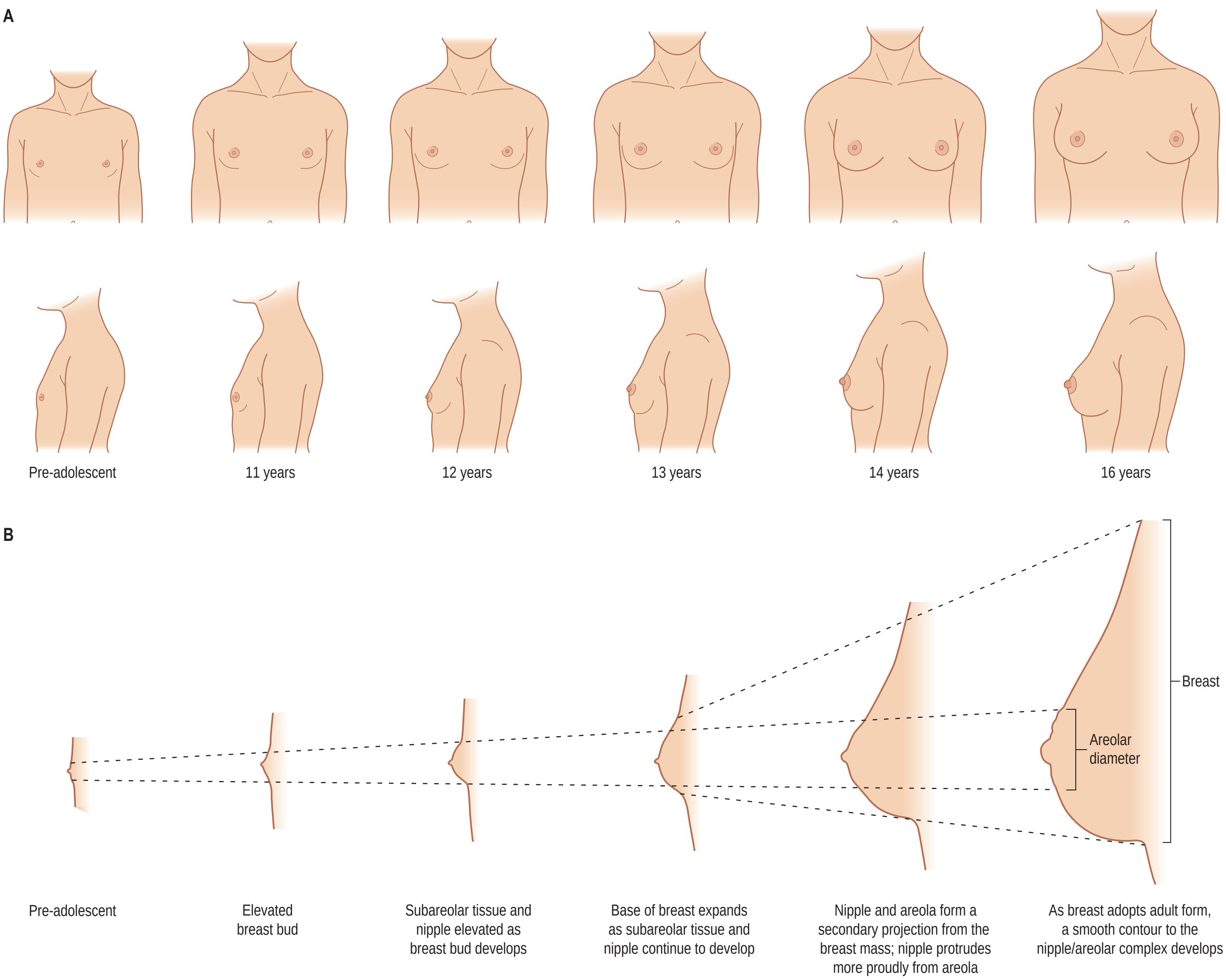

Fig. 53.29 Pre- and postpubertal development and structure of the female breast, demonstrating changes in the contour of the breast.

are responsible for the regression of the alveolar–ductal system: a reduction in epithelial cell size and a reduction in cell numbers mediated via apoptosis (p. 26). Gradually, the breast tissue reverts to its resting state. If another pregnancy occurs, the resting glandular tissue is reactivated and the process outlined above recurs. Up to the age of 50 years, increasing amounts of elastic tissue tend to be laid down around vessels and ducts (elastosis), and also in the stroma. Elastosis does not normally continue into later life.

Mammary stem cells Mammary stem cells (MaSCs) and distinct luminal progenitor cell types have been prospectively isolated from adult mouse and human mammary tissue, implying the existence of an epithelial differentiation hierarchy in the mammary gland. For further reading on the role that MaSCs may play in the postnatal mammary gland, see Rios et al (2014).

MALE BREAST

The male breast remains rudimentary throughout life. It is formed of small ducts (without lobules or alveoli) or solid cellular cords and a little supporting fibroadipose tissue (Ellis et al 1993). Slight temporary enlargement may occur in the newborn, reflecting the influence of maternal hormones, and again at puberty. The areola is well developed, although limited in area, and the nipple is relatively small. It is usually stated that the ducts do not extend beyond the areola in a male breast but glandular tissue can be more extensive.

Gynaecomastia

Gynaecomastia is a benign proliferation of subareolar breast tissue in the male. It may be unilateral or bilateral and of varying severity (Simon et al 1973) and symptomatology. Pain or tenderness, thought to be due to accumulation of fluid in the glandular ducts, may be a presenting feature. Histological changes are fibrosis and subsequent hyalinization. The aetiology in the majority of cases is idiopathic. Broadly speaking, causes may be either physiological, reflecting imbalances of sex hormones in neonates, at puberty and in senescence, or pathological, reflecting hypogonadal or hyperoestrogenic states. Acquired causes include metabolic imbalances, endocrine and neoplastic conditions, and drug-induced gynaecomastia. In the adult male, unilateral gynaecomastia should be investigated to exclude male breast cancer.

INTERVENTIONAL ACCESS TO THORACIC VISCERA

Thoracocentesis (pleural aspiration) Thoracocentesis or pleural aspiration is an essential step in the assessment of pleural effusions. A chest radiograph will confirm the location and extent of the effusion and clinical examination will identify the best position for aspiration, the posterior mid-scapular line being a common site. The skin of the desired interspace is cleaned and anaesthetized, and the aspiration needle is inserted at the lower margin of the interspace because the posterior intercostal vessels run mid-interspace. After appropriate local analgesia has been applied, the needle is carefully advanced in a perpendicular direction in the lower portion of the interspace until it enters the pleural space. More complex or small effusions should be aspirated under ultrasound guidance.

Needle thoracocentesis Needle thoracocentesis is performed when a life-threatening tension pneumothorax is suspected. A needle is inserted into the second intercostal space in the mid-clavicular line on the side of the tension pneumothorax, with the patient in an erect

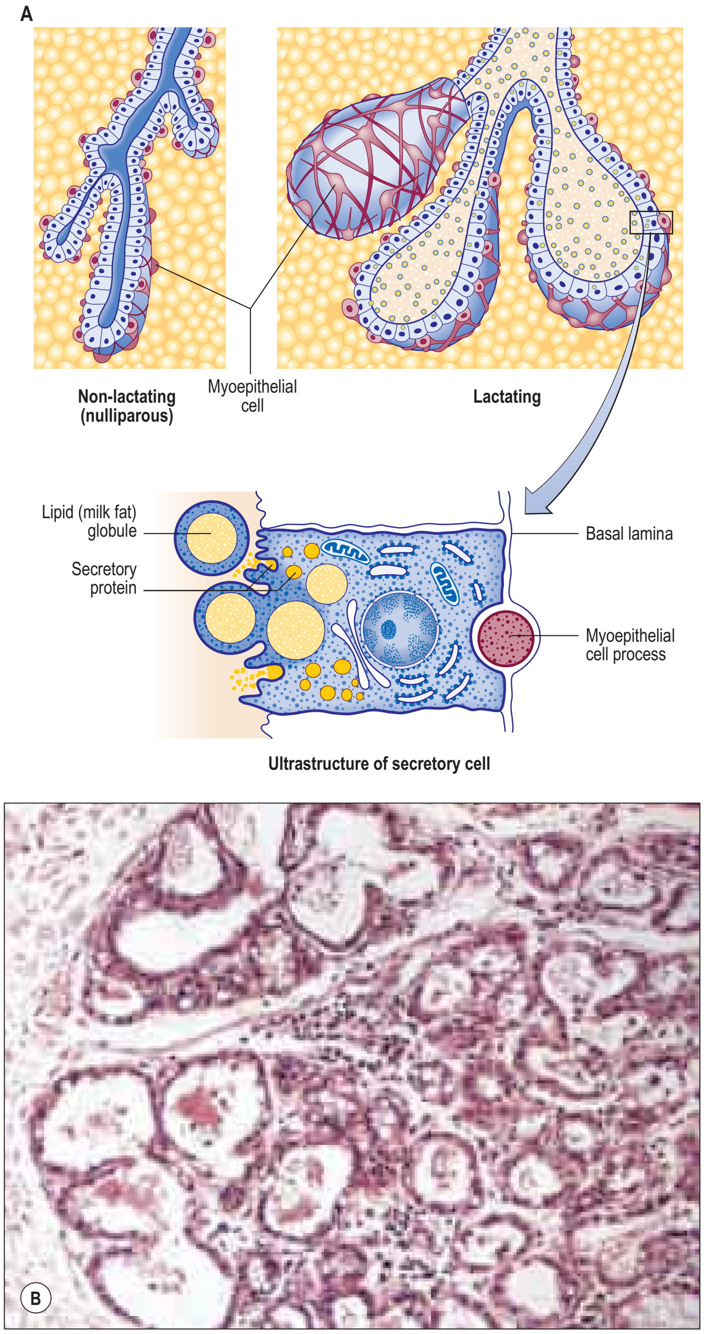

Fig. 53.30 The microstructure of breast epithelium. **A,** Note that the myoepithelial process is actually about half the relative size of that shown in the lower diagram. **B,** The peripheral part of a lactating breast lobule enclosed by a connective tissue septum (left). The alveoli are distended by milk secretion. Milk protein appears as eosinophilic material in the lumen and milk fat as pale cytoplasmic vacuoles in the flattened alveolar epithelium. Intralobular connective tissue between the alveoli contains a prominent lymphocytic infiltration, including plasma cells secreting IgA.

position. A sudden escape of air is heard when the needle enters the parietal pleura. A chest tube must be inserted after this procedure.

Chest drain insertion The insertion site for a chest drain is usually the fifth intercostal space, just anterior to the mid-axillary line on the affected side. A 2 cm horizontal incision is followed by blunt dissection through the subcutaneous tissues to the top of the rib. The parietal pleura is punctured with the tip of a clamp, and a gloved finger is inserted into the pleural space to free up any adhesions. The chest drain (thoracostomy tube) is then inserted into the pleural space and attached to an underwater sealed container placed below the level of the lungs; the water level rises and falls in the tube with ventilation.

Pericardiocentesis Pericardiocentesis is performed to aspirate a pericardial effusion or, in an emergency, to decompress a cardiac tamponade, where pressure from blood in the pericardial space prevents the heart chambers from filling during the cardiac cycle, seriously impairing cardiac output.

Pericardial puncture can be performed in either the fifth or sixth left intercostal space near the sternum (to avoid the internal thoracic artery) or at the left costoxiphoid angle. The needle is passed 1–2 cm to the left of the costoxiphoid angle at 45° to the skin, and then up and backwards towards the tip of the scapula until it enters the pericardial sac.

Placement of electrocardiograph (ECG) leads

Available with the *Gray's Anatomy* e-book

Thoracotomy incisions

Available with the *Gray's Anatomy* e-book

Sternotomy

Available with the *Gray's Anatomy* e-book

Axillary thoracotomy

Available with the *Gray's Anatomy* e-book

Thoracoscopic access

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Bonus e-book images

Fig. 53.7 Pectus excavatum.

Fig. 53.17 The internal thoracic vessels prepared for microanastomosis beneath the third costal cartilage.

Fig. 53.20 A,B, Pre- and post-embolization images after selective catheterization of the right eleventh posterior intercostal artery.

Fig. 53.26 A CT angiogram showing the deep inferior epigastric artery and a perforator emerging from the rectus sheath.

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The 12-lead ECG provides three-dimensional information on the electrical activity of the heart. The limb leads provide information about the electrical activity in the frontal plane. They are placed on the left and right wrists and the left foot; the right foot acts as a neutral grounding point. The chest leads provide information about the electrical activity in the horizontal plane and are placed as follows: V1, right fourth intercostal space, parasternal position; V2, left fourth intercostal space, parasternal position; V3, midpoint of V2 and V4 on the left; V4, fifth intercostal space, mid-clavicular line on the left; V5, fifth intercostal space, anterior axillary line on the left; and V6, fifth intercostal space, mid-axillary line on the left.

Thoracotomy incisions may be posterolateral or anterolateral, or involve a transverse thoracosternotomy.

Posterolateral incision A posterolateral incision is most commonly used in thoracic surgery for unilateral pulmonary resections, bullectomy, unilateral lung volume reduction surgery, chest wall resection and oesophageal surgery (Fry 2000). The patient is placed in a lateral decubitus position with adequate support of the elbow, axilla and knee with padding. The standard approach is via an incision from the anterior axillary line, which curves about 4 cm below the tip of the scapula and then vertically between the posterior midline and medial edge of the scapula. The incision is usually extended to the level of the spine of the scapula. Overall, the incision forms an S-shape in the fifth intercostal space. The sixth or seventh intercostal space is used in oesophageal surgery.

The lower portions of trapezius and latissimus dorsi are divided. Serratus anterior is retracted, and may be divided in a high thoracotomy. The costal muscle and pleura are dissected along the inferior margin of the intercostal space to avoid damaging the neurovascular bundle. A small section of rib is removed at the costovertebral angle to reduce the risk of fracture, particularly in patients older than 40 years. This technique provides good access to the thoracic contents; the main problem is postoperative pain as a consequence of intraoperative musculoskeletal traction.

Anterolateral incision The patient is placed in the supine position, with the arms by the sides. A roll is placed vertically under the back and hips in order to raise the operative side by approximately 45°. The incision is from the mid-axillary line over the fifth intercostal space along the inframammary fold, and curves upwards parasternally. The pectoral muscles are divided, and subsequent access to the thorax is similar to that used in the posterolateral approach. However, access is limited, and may be improved by dividing costal cartilages.

'Clam shell' The transverse thoracosternotomy is known as the 'clam shell' incision. It provides excellent exposure to both sides of the chest and is therefore used in bilateral lung transplantation and in lung volume reduction surgery with bilateral lung resections. The patient is placed in the supine position with a roll vertically along the upper thoracic spine. Bilateral anterolateral incisions are made in the inframammary fold, and the sternum is transected. This allows the upper portion of the thorax to be displaced upwards with a rib-spreader; hence the name 'clam shell'. The main disadvantage of the clam shell procedure is the need to transect the sternum; even after careful repair with sternal wires, there is a risk of sternal instability.

Sternotomy is commonly needed in cardiac surgery. The patient is placed in the supine position with both arms extended by the side. A vertical incision is made in the midline from the suprasternal notch to a point just below the xiphoid process; the tissues around the manubrium and the xiphoid process are mobilized; and the pectoral fascia in the midline is incised. The sternum is split and its two edges retracted; it is subsequently closed using interosseous wire sutures.

The patient is placed in the lateral decubitus position, with arms abducted at 90° and supported on an arm rest. The incision is based along the desired intercostal space; for upper thoracic lesions, this is the second or third space. Latissimus dorsi is elevated and retracted, whereas serratus anterior is divided in the direction of its fibres. The anterior aspect of serratus anterior is divided to expose the intercostal muscles, which are divided in turn. The overall size of the incision is limited and it provides good access to the upper thorax. Postoperative pain is less than with some other approaches, but the long thoracic nerve (nerve of Bell) may be damaged if serratus anterior is divided too posteriorly.

Occasionally, video-assisted thoracoscopic surgery is required to assess the mediastinum. The thoracoscope is usually introduced via the fifth intercostal space in the mid-clavicular line, with additional ports at the third and sixth intercostal spaces to assess the anterior mediastinum. To assess the posterior mediastinum, the thoracoscope is inserted into the seventh intercostal space in the mid-clavicular line.

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